Community Development – Zoning Division
John Pederson – Division Manager

ZONING CASE

Z-5-2026

SITE BACKGROUND

Applicant	The Revive Land Group, LLC
Phone	Contact Zoning Office
Email	Contact Zoning Office
Representative Contact	J. Kevin Moore
Phone	Contact Zoning Office
Email	Contact Zoning Office
Titleholder	Estate of Lucille Garrett a/k/a Leathy Lucille Garrett a/k/a Leathy Lucille Price Garrett and William D. Brown
Property Location	Located on the east and west sides of Ernest Barrett Parkway and the western terminus of Old Horseshoe Bend Road
Address	None assigned
Access to Property	Ernest Barrett Parkway and Old Horseshoe Bend Road

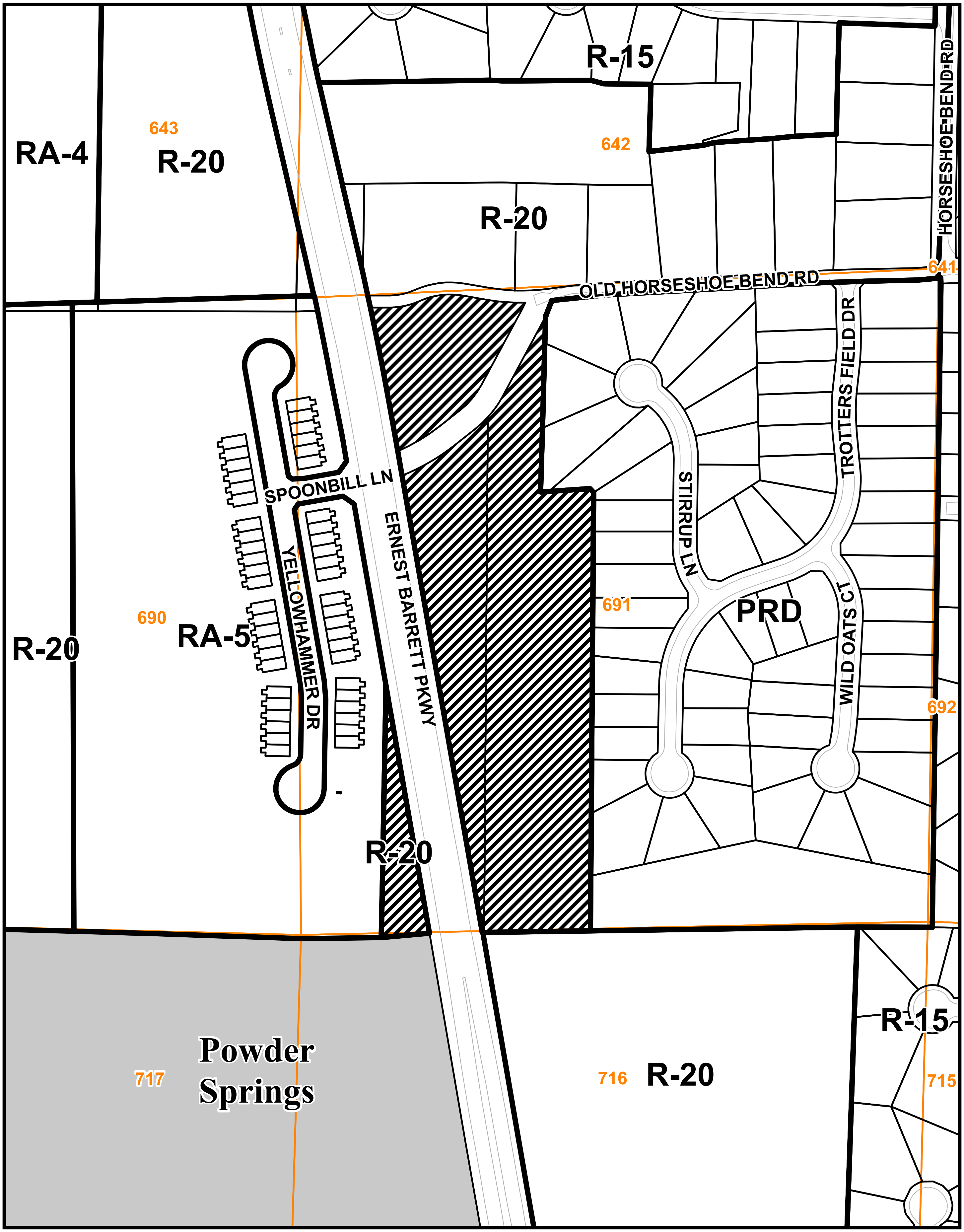
QUICK FACTS

Commission District	4 - Sheffield
Current Zoning	R-20
Current Use of Property	Vacant
Proposed Zoning	RA-5
Proposed Use	Single-family community
Future Land Use	MDR
Site Acreage	10.17
District	19
Land Lot	691
Parcel #	19069100010, 19069100070
Taxes Paid	Yes

FINAL ZONING STAFF RECOMMENDATIONS

Click here to enter text.

Z-5 2026 Map

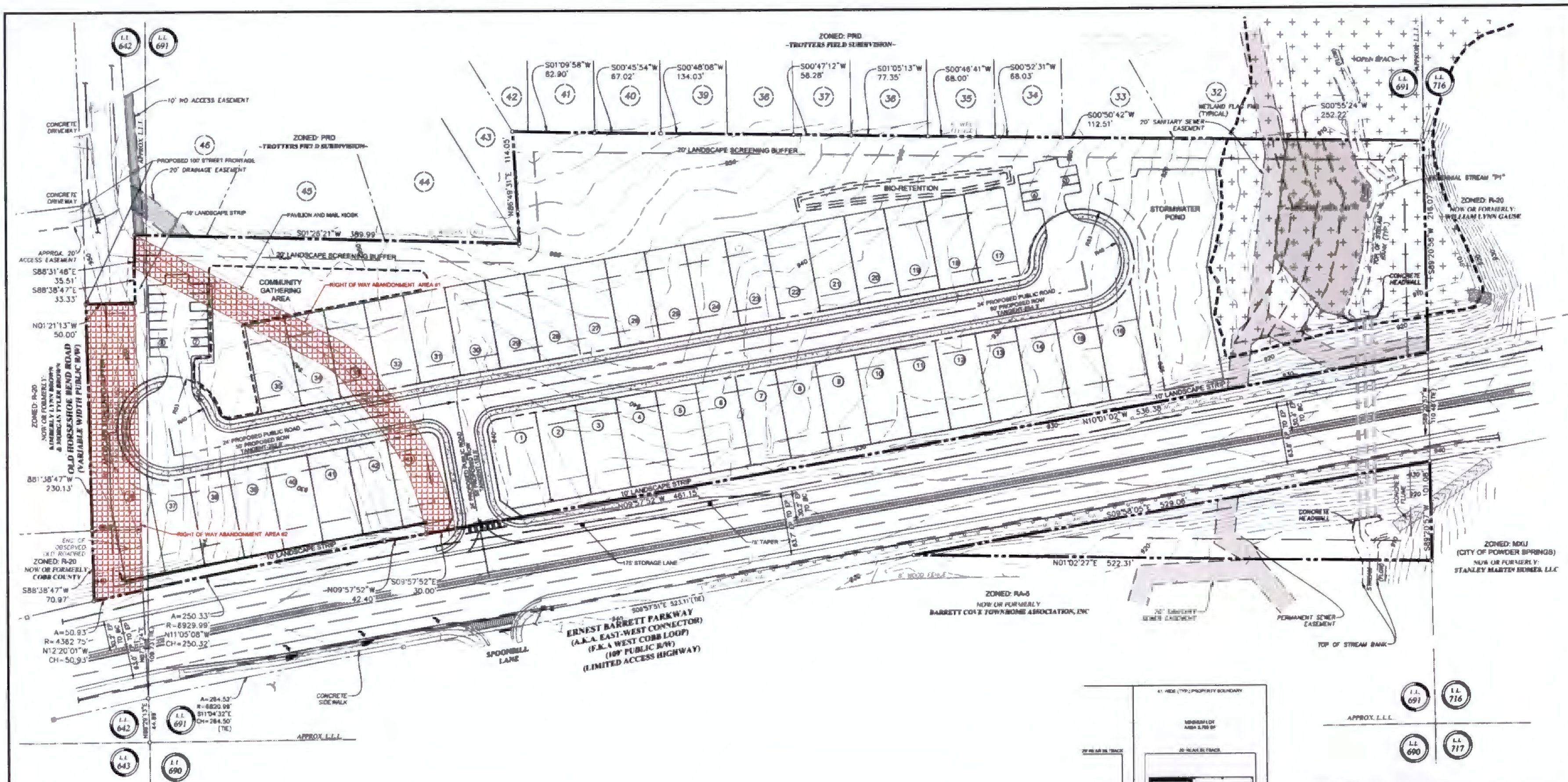


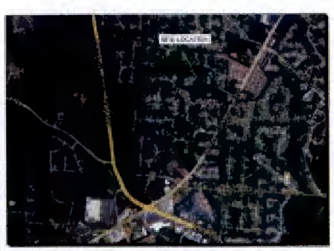
This map is provided for display and planning purposes only. It is not meant to be a legal description.

0 230 460 Feet

Zoning Boundary

City Boundary





SITE LOCATION MAP

NOT TO SCALE

GEORGIA811

Know what's below. Call before you dig.

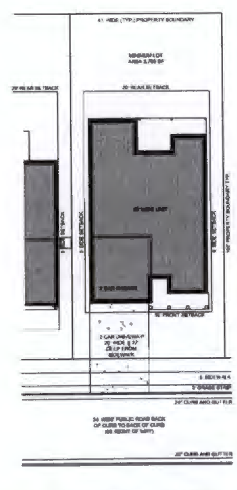
24 HOUR CONTACT:
NEVILLE ALLISON
678-223-8978

GENERAL NOTES:

1. STREAMS AND WETLANDS ARE LOCATED ON THE SITE
2. PROPERTY IS LOCATED WITHIN A SPECIAL FLOOD HAZARD AREA ACCORDING TO FEDERAL EMERGENCY MANAGEMENT AGENCY (FEMA), AS SHOWN ON FLOOD INSURANCE RATE MAP PANEL [illegible], WITH A DATE OF IDENTIFICATION OF MARCH 4, 2013 FOR COMMUNITY NO. 13067C, COBB COUNTY, GEORGIA, WHICH IS THE CURRENT FLOOD INSURANCE RATE MAP FOR THE COMMUNITY IN WHICH SAID PREMISES IS LOCATED
3. POWER AND GAS EASEMENTS ARE LOCATED ON THE SITE IN THE EXISTING RIGHT OF WAY
4. NO CEMETERIES ARE LOCATED ON THE SITE
5. SPEED LIMIT ON ERNEST BARRETT PARKWAY IS 45 MPH
6. WATER UTILITY PROVIDER IS COBB COUNTY
7. SANITARY SEWER PROVIDER IS COBB COUNTY
8. SIZE AND LOCATION OF PROPOSED STRUCTURES AS INDICATED ON SITE PLAN

0 50 100 150 feet

1" = 50'-0"



TYPICAL LOT LAYOUT

SITE AREA	10.171 ACRES OR 443,030 SF
FLOOD PLAIN	0.960 ACRES OR 41,845 SF (½ = 0.478 AC)
NET SITE AREA	9.693 ACRES OF 422,216 SF
EXISTING ZONING	R-20
PROPOSED ZONING	RA-6
MINIMUM LOT AREA	7,000 SF (PROPOSED 3,700 SF)
MIN./MAX. ACERAGE REQUIRED	[illegible] SF/20 ACRES
MAXIMUM DENSITY	6 UNITS PER ACRE
MINIMUM FLOOR AREA	900 SF
MINIMUM LOT WIDTH AT FRONT SETBACK	70 FEET/50 FEET (PROPOSED 41 FEET)
FRONT YARD SETBACK ARTERIAL STREET	50 FEET (N/A)
FRONT YARD SETBACK INTERIOR STREET	20 FEET (PROPOSED 10 FEET)
MAJOR SIDE YARD SETBACK	20 FEET (PROPOSED 10 FEET)
SIDE SETBACK INTERIOR	5 FEET
REAR SETBACK ARTERIAL STREET	40 FEET (PROPOSED 30 FEET)
REAR SETBACK INTERIOR	30 FEET (PROPOSED 20 FEET)
MINIMUM FLOOR AREA	900 SF
BUILDING HEIGHT	35 FEET
MAXIMUM COVERAGE	40% OR 1,[illegible] SF (PROPOSED 40% OR 1,[illegible] SF)
MAXIMUM IMPERVIOUS AREA	35% OR 155,[illegible] SF
PROPOSED UNITS	43 UNITS
PROPOSED DENSITY	4.44 UNITS PER ACRE
GUEST PARKING SPACES (0.5 SPACES PER UNIT)	22 SPACES OR 0.51 SPACES PER UNIT

THE REVIVE LAND GROUP

SOURCE. ENTITLE. REVIVE.

ONE ALLIANCE CENTER
3500 LENOX ROAD
SUITE 625
ATLANTA, GEORGIA 30326
O: (678) 223-8978
WWW.THEREVIVELANDGROUP.COM

PROJECT

SOMERSET AT MACEDONIA

A MASTER PLANNED RESIDENTIAL DEVELOPMENT

OLD HORSESHOE BEND ROAD AND BARRETT PARKWAY
MARIETTA, GEORGIA 30064

COBB COUNTY GEORGIA

LAND LOT: 691
DISTRICT: 19TH

CLIENT

THE REVIVE LAND GROUP

SOURCE. ENTITLE. REVIVE.

ONE ALLIANCE CENTER
3500 LENOX ROAD
SUITE 625
ATLANTA, GEORGIA 30326
O: (678) 223-8978
THEREVIVELANDGROUP.COM

REVISION | DATE

PROJECT NUMBER

25021

PRELIMINARY
NOT TO BE RELEASED FOR CONSTRUCTION

NORTH

DATE

NOVEMBER 20, 2025

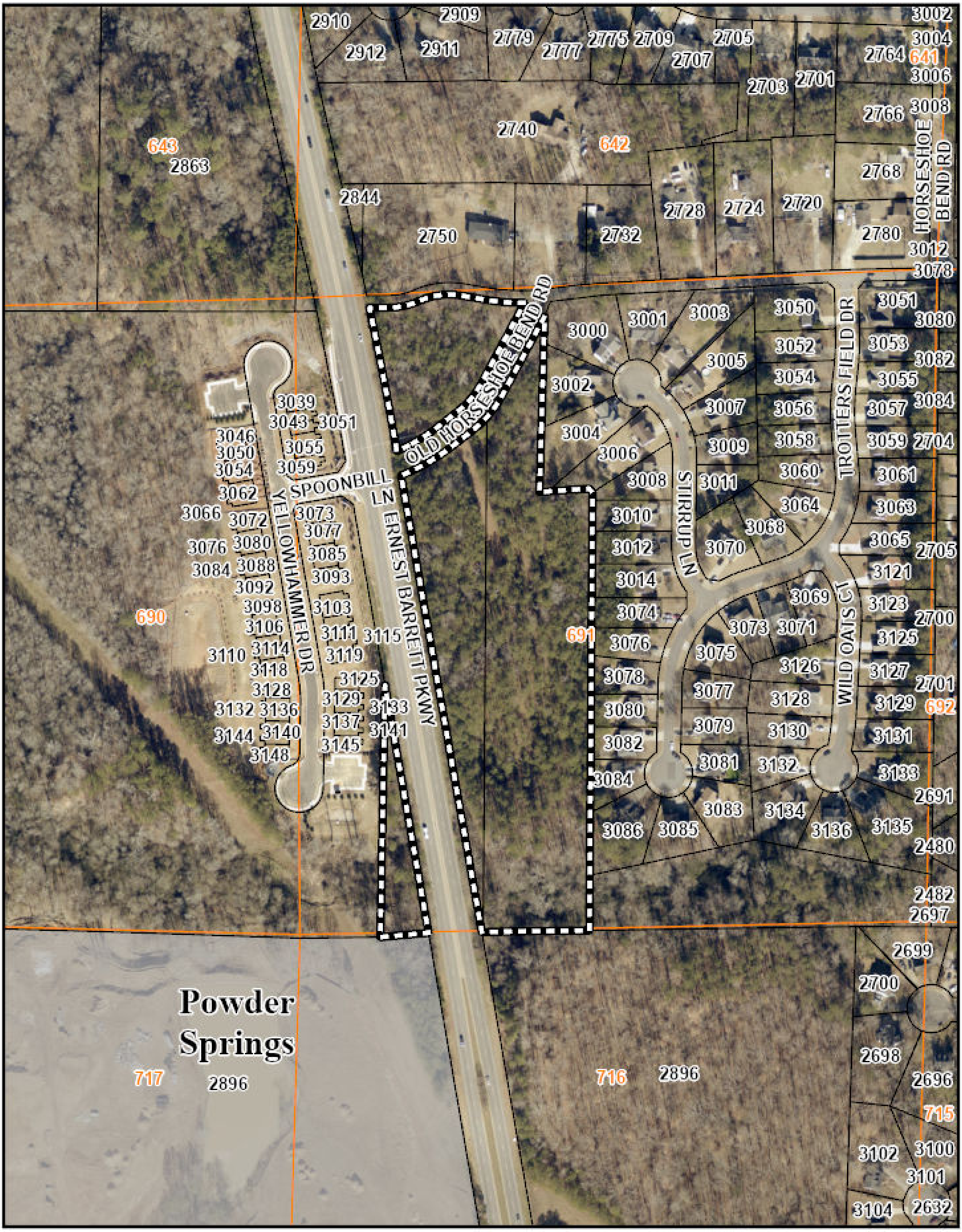
TITLE

CONCEPTUAL SITE PLAN

SHEET

Z-01

Z-5 2026 Aerial Map



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0 230 460 Feet

Zoning Boundary

City Boundary

Application No. Z-   (2026)

Hearing Dates: PC - 02/03/2026
BOC - 02/17/2026

Summary of Intent for Rezoning*

Part 1. Residential Rezoning Information (attach additional information if needed)

a) **Proposed unit square-footage(s):** Minimum 1,600 square feet and greater

b) **Proposed building architecture:** Traditional/Craftsman - Fronts of brick, stone, shake, board and batten, cementitious lap siding

c) **List all requested variances:** See attached Listing of Variances

Part 2. Non-residential Rezoning Information (attach additional information if needed)

a) **Proposed use(s):** Not Applicable.

b) **Proposed building architecture:** Not Applicable.

c) **Proposed hours/days of operation:** Not Applicable.

d) **List all requested variances:**

Part 3. Other Pertinent Information (List or attach additional information if needed)

Part 4. Is any of the property included on the proposed site plan owned by the Local, State, or Federal Government? **(Please list all Right-of-Ways, Government owned lots, County owned parcels and/or remnants, etc., and attach a plat clearly showing where these properties are located).** Yes. Portions of two Cobb County rights-of-way are proposed for abandonment, as shown on submitted Site Plan.

*Applicant specifically reserves the right to amend any information set forth in this Summary of Intent for Rezoning, or any portion of the Application for Rezoning or documentation attached thereto, at any time during the rezoning process.

ATTACHMENT TO SUMMARY OF INTENT FOR REZONING

Application No.: Z-__________ (2026)
Hearing Dates: February 3, 2026
February 17, 2026

Applicant: The Revive Land Group, LLC
Titleholder: Estate of Lucille Garrett
(a/k/a Leathy Lucille Garrett;
a/k/a Leathy Lucille Price Garrett)

Tax Parcel Identification No.: 19069100010

Part 1. Residential Rezoning Information

(c) List all requested variances:

i) Sec. 134-201.2.(4)a. Minimum lot size: 7,000 sf. Variance request for minimum lot size: 3,700 sf;
ii) Sec. 134-201.2.(4)b. Minimum lot width at front setback line: 70 feet; cul-de-sac or interior lot, 50 feet. Variance request for minimum lot width of 41 feet;
iii) Sec. 134-201.2.(4)c. Minimum width between buildings: 15 feet. Variance request for minimum width between buildings of 10 feet;
iv) Sec. 134-201.2.(4)d. Minimum public road frontage: 70 feet; cul-de-sac 35 feet; 50 feet minimum public road frontage if interior to development. Variance request for minimum public road frontage of 41 feet;
v) Sec. 134-201.2.(4)e. Minimum building setbacks: minimum front yard setback: 20 feet. Variance request for minimum front yard setback of 15 feet;
vi) Sec. 134-201.2.(4)e. Minimum building setbacks: minimum major side yard setback: 20 feet. Variance request for minimum front yard setback of 15 feet.
vii) Sec. 134-201.2.(4)e. Minimum building setbacks: minimum rear yard setback on an arterial street: 40 feet. Variance request for minimum rear yard setback on an arterial street of 30 feet;
viii) Sec. 134-201.2.(4)e. Minimum building setbacks: minimum rear yard setback: 30 feet. Variance request for minimum front yard setback of 20 feet; and
ix) Sec. 134-191. Summary of bulk regulations: maximum coverage: 35 percent. Variance request for maximum coverage of 45percent.

Cobb County Development Standards Sec 402.7.1

x) A minimum 50 feet straight line distance is required between residential driveways. Request a minimum 12 feet straight line distance between residential driveways; and
xi) A minimum 50 feet straight line distance is required between the intersection radius and the edge of residential driveway aprons. Request a minimum 2 feet straight line distance between the intersection radius and the edge of the residential driveway apron.

Traffic Impact Study
for
Proposed Somerset at Macedonia Residential Development on Ernest W Barrett Parkway

Cobb County, Georgia

GEORGIA
REGISTERED
NO. 22518
PROFESSIONAL
ENGINEER
ABDUL KAREEM AMER

Prepared for:

The Revive Land Group
3500 Lenox Road, Suite 625
Atlanta, GA 30326

Prepared By:

A&R Engineering Inc.
2160 Kingston Court, Suite O
Marietta, GA 30067
Tel: (770) 690-9255 Fax: (770) 690-9210
www.areng.com

November 17, 2025
A & R Project # 25-196

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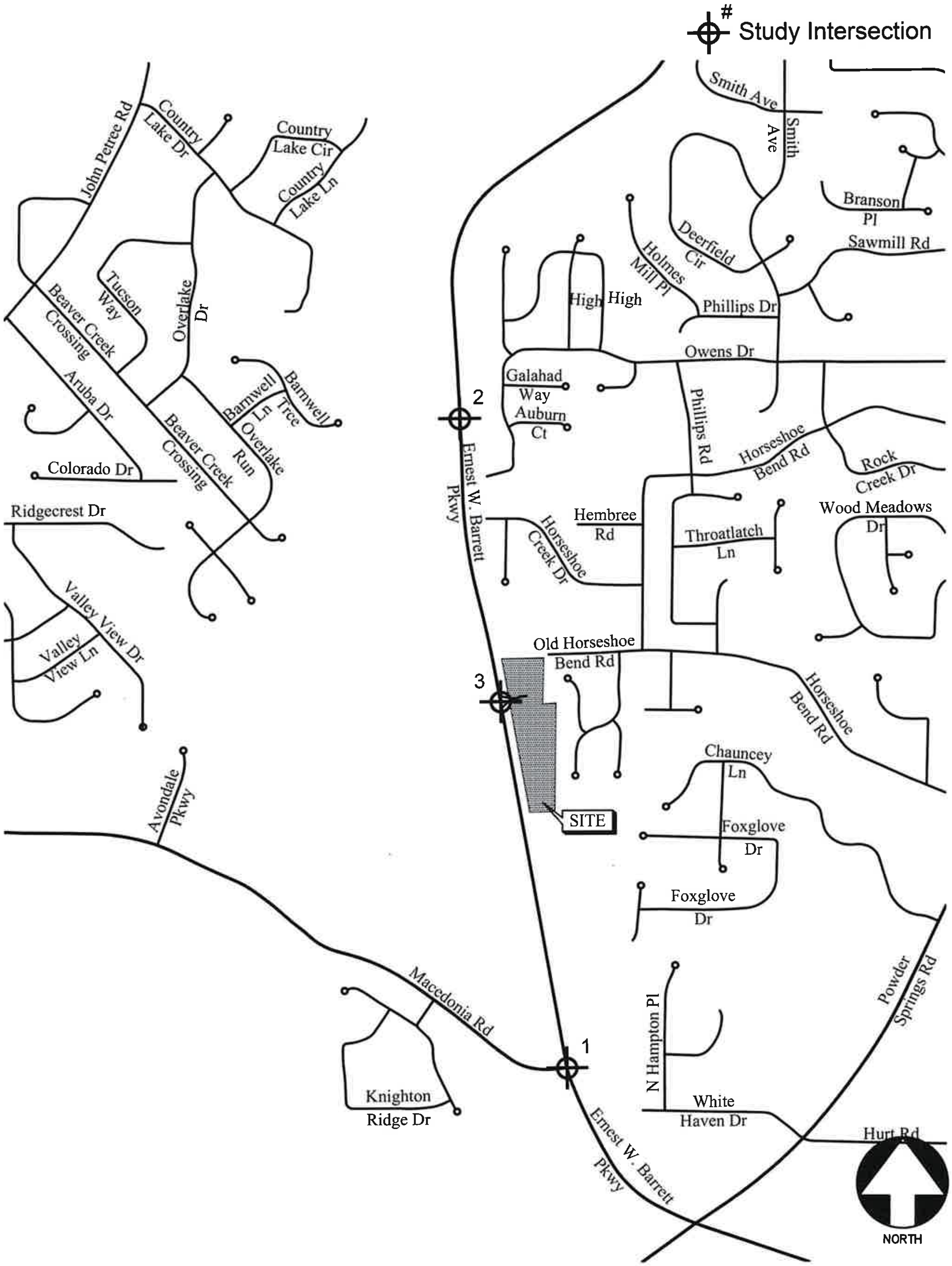
1.0 INTRODUCTION

The purpose of this study is to determine the traffic impact from the proposed Somerset at Macedonia residential development that will be located on the east side of Ernest W Barrett Parkway about 3,000 feet north of Macedonia Road in Cobb County. The traffic analysis includes evaluation of the current operations as well as future conditions including traffic generated by the proposed development. The development will consist of 43 single-family detached houses and proposes one right-in/right-out driveway on Ernest W Barrett Parkway.

The AM and PM peak hours have been analyzed in this study. In addition to the site access point, this study includes the evaluation of traffic operations at the intersections of:

1. Ernest W Barrett Parkway at Macedonia Road
2. Ernest W Barrett Parkway at Median Break (approximately 0.45 miles to the north of the proposed site driveway location)

Recommendations to improve traffic operations have been identified as appropriate and are discussed in detail in the following sections of the report. The location of the development and the surrounding roadway network are shown in Figure 1.



LOCATION MAP

FIGURE 1

A&R Engineering Inc.

2.0 EXISTING FACILITIES / CONDITIONS

2.1 Roadway Facilities

The following is a brief description of each of the roadway facilities located in proximity to the site:

2.1.1 Ernest W Barrett Parkway

Ernest W Barrett Parkway is a north-south, four-lane, median divided roadway with a posted speed limit of 45 mph in the vicinity of the site. Georgia Department of Transportation (GDOT) traffic counts (Station ID: 067-0657) indicate that the daily traffic volume on Ernest W Barrett Parkway in 2024 was 21,500 vehicles per day northwest of Macedonia Road. Cobb County DOT classifies Ernest W Barrett Parkway as an arterial roadway.

2.1.2 Macedonia Road

Macedonia Road is an east-west, two-lane, undivided roadway with a two way left turn lane and posted speed limit of 35 mph in the vicinity of the site. Georgia Department of Transportation (GDOT) traffic counts (Station ID: 067-8836) indicate that the daily traffic volume on Macedonia Road in 2024 was 5,660 vehicles per day southwest of Hopkins Road. Cobb County DOT classifies Macedonia Road as a major collector roadway.

3.0 Study Methodology

In this study, the methodology used for evaluating traffic operations at each of the subject intersections is based on the criteria set forth in the Transportation Research Board's Highway Capacity Manual, 6th edition (HCM 6). Synchro software, which utilizes the HCM methodology, was used for the analysis. The following is a description of the methodology employed for the analysis of unsignalized and signalized intersections.

3.1 Unsignalized Intersections

For unsignalized intersections controlled by a stop sign on minor streets, the level of service (LOS) for motor vehicles with controlled movements is determined by the computed control delay according to the thresholds stated in Table 1 below. LOS is determined for each minor street movement (or shared movement), as well as major street left turns. LOS is not defined for the intersection as a whole or for major street approaches. The LOS of any controlled movement which experiences a volume to capacity ratio greater than 1 is designated as "F" regardless of the control delay.

Control delay for unsignalized intersections includes initial deceleration delay, queue move-up time, stopped delay, and final acceleration delay. Several factors affect the control delay for unsignalized intersections, such as the availability and distribution of gaps in the conflicting traffic stream, critical gaps, and follow-up time for a vehicle in the queue.

Level of service is assigned a letter designation from "A" through "F". Level of service "A" indicates excellent operations with little delay to motorists, while level of service "F" exists when there are insufficient gaps of acceptable size to allow vehicles on the side street to cross the main road without experiencing long delays.

Table 1 – Level of Service Criteria for Unsignalized Intersections

Control Delay (sec/vehicle)	LOS by Volume-to-Capacity Ratio*	
	v/c ≤ 1.0	**v/c > 1.0**
≤ 10	A	F
> 10 and ≤ 15	B	F
> 15 and ≤ 25	C	F
> 25 and ≤ 35	D	F
> 35 and ≤ 50	E	F
> 50	F	F

*The LOS criteria apply to each lane on a given approach and to each approach on the minor street. LOS is not calculated for major-street approaches or for the intersection.

Source: Highway Capacity Manual, 6th edition, Exhibit 20-2 *LOS Criteria: Motorized Vehicle Mode*

3.2 Signalized Intersections

According to HCM procedures, LOS can be calculated for the entire intersection, each intersection approach, and each lane group. HCM uses control delay alone to characterize LOS for the entire intersection or an approach. Control delay per vehicle is composed of initial deceleration delay, queue move-up time, stopped delay, and final acceleration delay. Both control delay and volume-to-capacity ratio are used to characterize LOS for a lane group. A volume-to-capacity ratio of greater than 1.0 for a lane group indicates failure from capacity perspective. Therefore, such a lane group is assigned LOS F regardless of the amount of control delay.

Table 2 below summarizes the LOS criteria from HCM for motorized vehicles at signalized intersection.

TABLE 2 – LEVEL OF SERVICE CRITERIA FOR SIGNALIZED INTERSECTIONS

Control Delay (sec/vehicle) *	LOS for Lane Group by Volume-to-Capacity Ratio*	
	v/c ≤ 1.0	**v/c > 1.0**
≤ 10	A	F
> 10 and ≤ 20	B	F
> 20 and ≤ 35	C	F
> 35 and ≤ 55	D	F
> 55 and ≤ 80	E	F
> 80	F	F

*For approach-based and intersection wide assessments, LOS is defined solely by control delay

Source: Highway Capacity Manual, 6th edition, Exhibit 19-8 *LOS Criteria: Motorized Vehicle Mode*

LOS A is typically assigned when the volume-to-capacity (v/c) ratio is low and either progression is exceptionally favourable, or the cycle length is very short. LOS B is typically assigned when the v/c ratio is low and either progression is highly favorable, or the cycle length is short. However, more vehicles are stopped than with LOS A. LOS C is typically assigned when progression is favorable, or the cycle length is moderate. Individual *cycle failures* (one or more queued vehicles are not able to depart because of insufficient capacity during the cycle) may begin to appear at this level. Many vehicles still pass through the intersection without stopping, but the number of vehicles stopping is significant. LOS D is typically assigned when the v/c ratio is high and either progression is ineffective, or the cycle length is long. There are many vehicle-stops and individual cycle failures are noticeable. LOS E is typically assigned when the v/c ratio is high, progression is very poor, the cycle length is long, and individual cycle failures are frequent. LOS F is typically assigned when the v/c ratio is very high, progression is very poor, the cycle length is long, and most cycles fail to clear the queue.

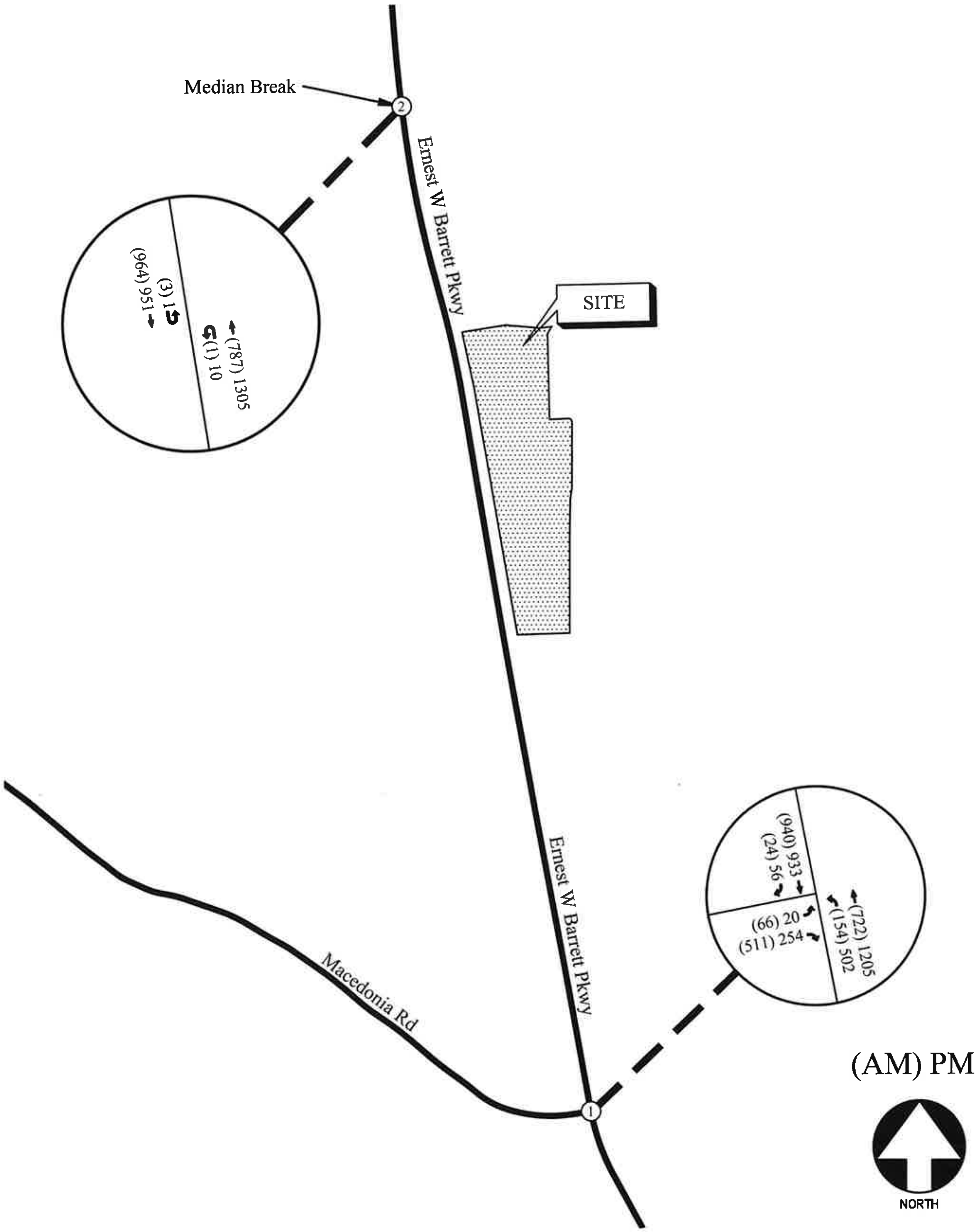
4.0 EXISTING 2025 TRAFFIC ANALYSIS

4.1 Existing Traffic Volumes

Existing traffic counts were obtained at the following study intersections:

1. Ernest W Barrett Parkway at Macedonia Road
2. Ernest W Barrett Parkway at Median Break

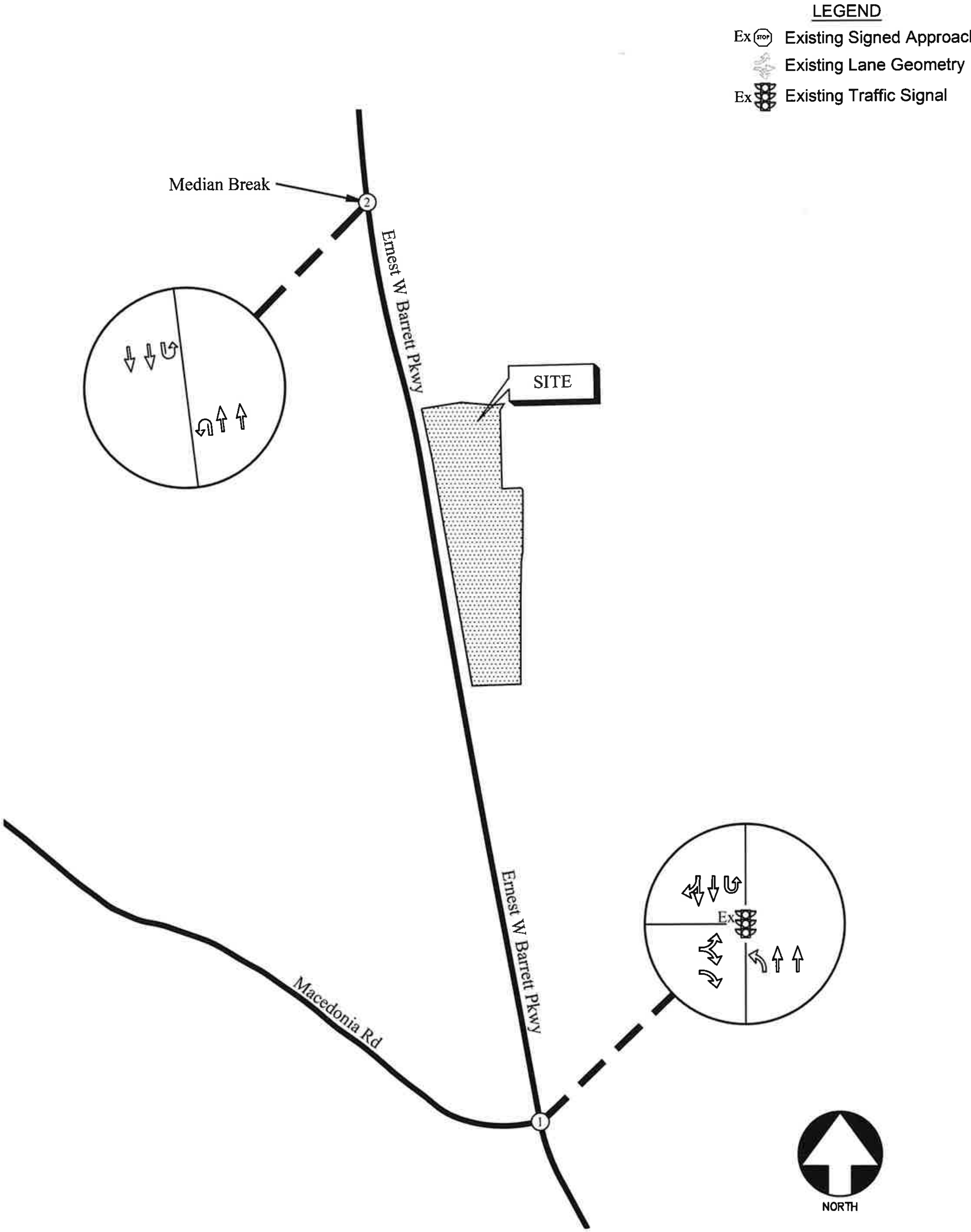
Turning movement counts were collected on Wednesday, October 22, 2025. All turning movement counts were recorded during the AM and PM peak hours between 7:00am to 9:00am and 4:00pm to 6:00pm, respectively. The four consecutive 15-minute interval volumes that summed to produce the highest volume at the intersections were then determined. These volumes make up the peak hour traffic volumes for the intersections and are shown in Figure 2. The existing traffic control and lane geometry for the intersections are shown in Figure 3.



EXISTING WEEKDAY PEAK-HOUR VOLUMES

FIGURE 2

A&R Engineering Inc.



EXISTING TRAFFIC CONTROL AND LANE GEOMETRY FIGURE 3

A&R Engineering Inc.

Figure 3 – Existing Traffic Control and Lane Geometry

4.2 Existing Traffic Operations

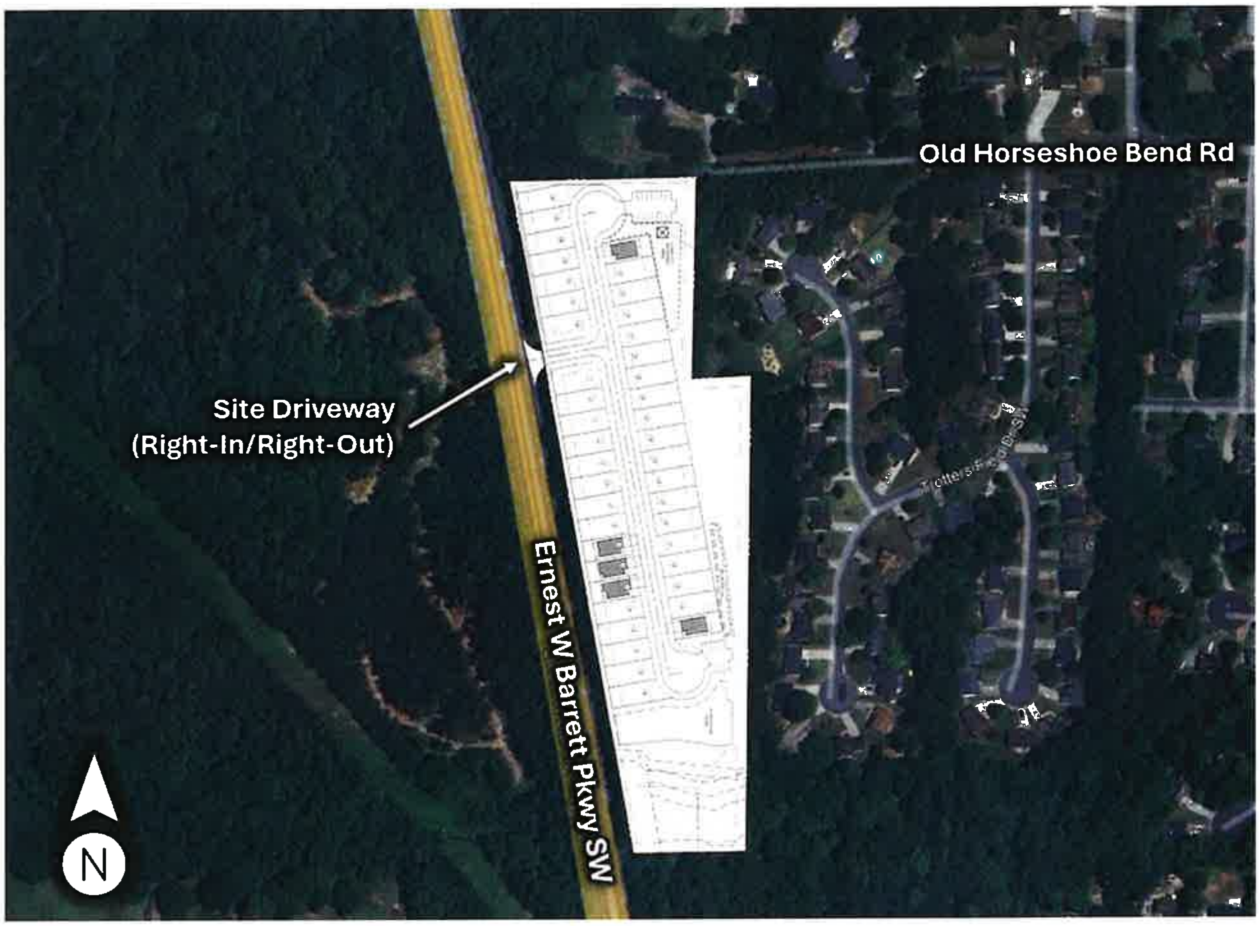
Existing 2025 traffic operations were analyzed at the study intersections in accordance with the HCM methodology. The results of the analyses are shown in Table 3.

<table>
<tr><th colspan="5">TABLE 3 – EXISTING INTERSECTION OPERATIONS</th></tr>
<tr><th colspan="2" rowspan="2">Intersection</th><th rowspan="2">Traffic Control</th><th colspan="2">LOS (Delay)</th></tr>
<tr><th>AM Peak Hour</th><th>PM Peak Hour</th></tr>
<tr><td>1</td><td><u>Ernest W Barrett Parkway @ Macedonia Road</u>
-Eastbound Approach
-Northbound Approach
-Southbound Approach</td><td>Signalized</td><td><u>C (23.0)</u>
D (54.9)
A (8.8)
B (16.6)</td><td><u>C (21.0)</u>
E (74.4)
B (15.2)
B (16.0)</td></tr>
<tr><td>2</td><td><u>Ernest W Barrett Parkway @ Median Break</u>
-Northbound U-Turn
-Southbound U-Turn</td><td>-</td><td>
A (0.0)
A (0.0)</td><td>
A (0.0)
A (0.0)</td></tr>
</table>

The results of the existing traffic operations analysis indicate that the signalized study intersection is operating at an overall level of service "B" in both the AM and PM peak hours.

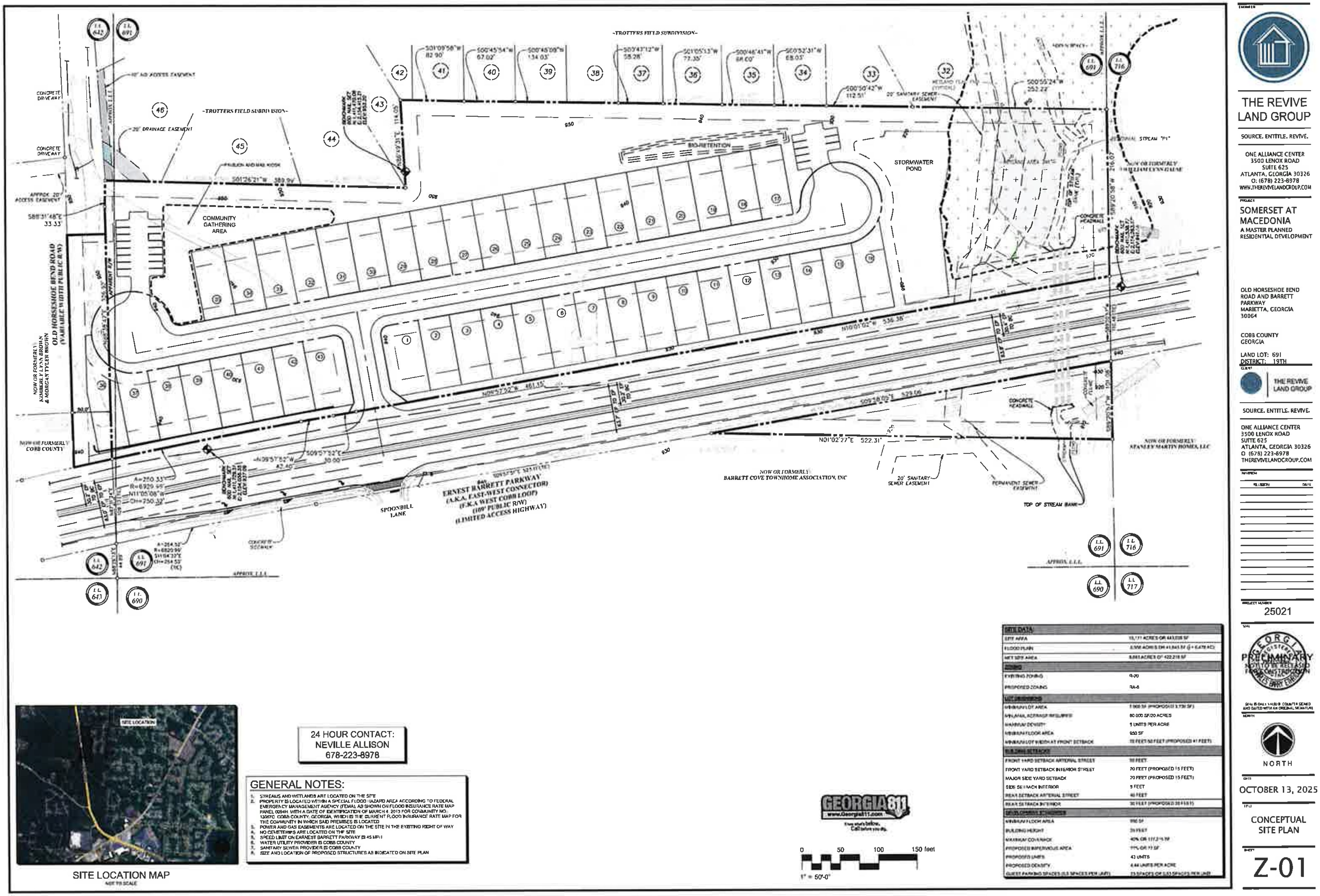
5.0 PROPOSED DEVELOPMENT

The proposed development is the Somerset at Macedonia residential development that will be located on the east side of Ernest W Barrett Parkway about 3,000 feet north of Macedonia Road in Cobb County. The development will consist of 43 single-family detached houses and proposes one right-in/right-out driveway on Ernest W Barrett Parkway.



A site plan is shown in Figure 4.

Figure 4 – Site Plan



5.1 Trip Generation

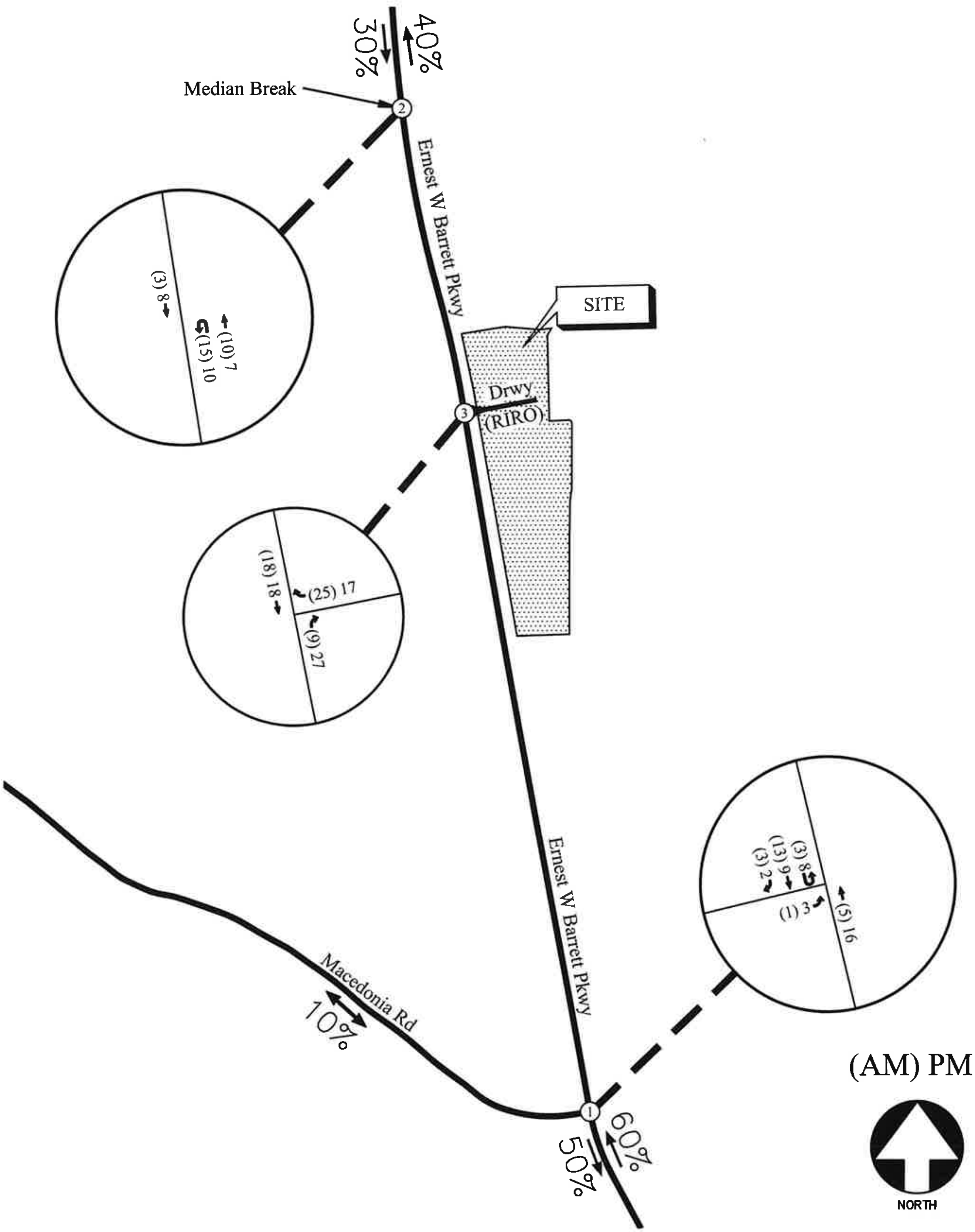
Trip generation estimates for the project were based on the rates and equations published in the 12th edition of the Institute of Transportation Engineers (ITE) Trip Generation Manual. This reference contains traffic volume count data collected at similar facilities nationwide. The trip generation was based on the following ITE Land Use category: *210 – Single-Family Detached Housing.* The calculated trip generation volumes for the proposed development are shown in Table 4.

TABLE 4 – TRIP GENERATION FOR PROPOSED DEVELOPMENT

Land Use	Size	AM Peak Hour			PM Peak Hour			24 Hour
		Enter	Exit	Total	Enter	Exit	Total	2-way
ITE 210 – Single-Family Detached Housing	43 Units	9	25	34	27	17	44	612

5.2 Trip Distribution

The trip distribution describes how traffic arrives and departs from the site. An overall trip distribution was developed for the site based on a review of the existing travel patterns in the area and the locations of major roadways and highways that will serve the development. The site-generated peak hour traffic volumes, shown in Table 4, were assigned to the study area intersections based on this distribution. The outer-leg distribution and AM and PM peak hour traffic volumes generated by the site are shown in Figure 5.



TRIP DISTRIBUTION AND NEW SITE-GENERATED WEEKDAY PEAK HOUR VOLUMES

FIGURE 5

A&R Engineering Inc.

6.0 Future 2027 Traffic Analysis

The future 2027 traffic operations are analyzed for the "Build" and "No-Build" conditions.

6.1 Future "No-Build" Conditions

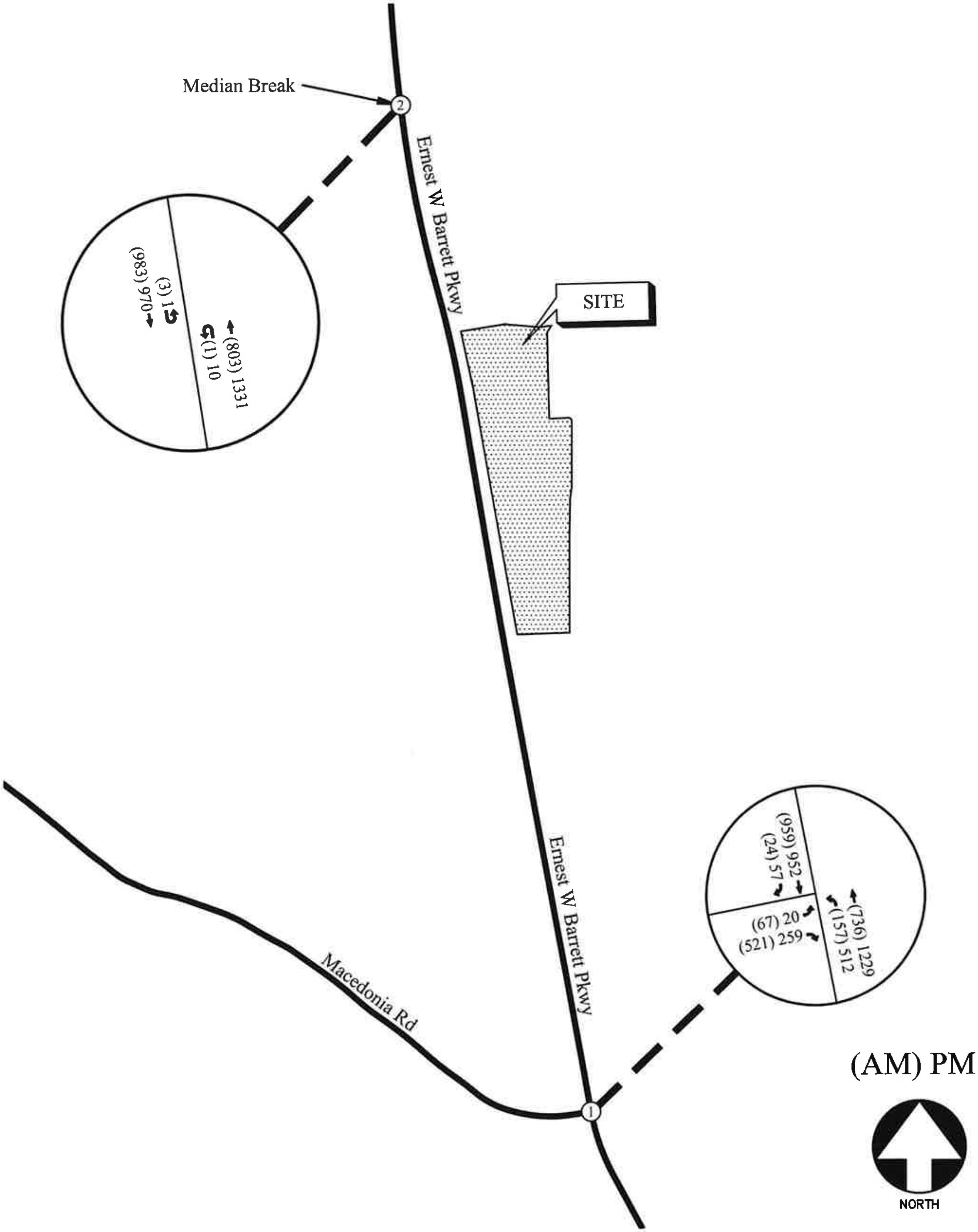
The "No-Build" (or background) conditions provide an assessment of how traffic will operate in the study horizon year without the study site being developed as proposed, with projected increases in through traffic volumes due to normal annual growth. The Future "No-Build" volumes consist of the existing traffic volumes (Figure 2) plus increases for annual growth of through traffic.

6.1.1 Annual Traffic Growth

In order to evaluate future traffic operations in this area, a projection of normal traffic growth was applied to the existing volumes. The Georgia Department of Transportation recorded average daily traffic volumes at several locations in the vicinity of the site. Reviewing the growth over the last three (2022-2024) years revealed growth of approximately 1% in the area. This growth factor was applied to the existing traffic volumes between collector and arterial roadways to estimate the future year traffic volumes prior to the addition of site-generated traffic. The resulting future "No-Build" volumes on the roadway are shown in Figure 6.

6.2 Future "Build" Conditions

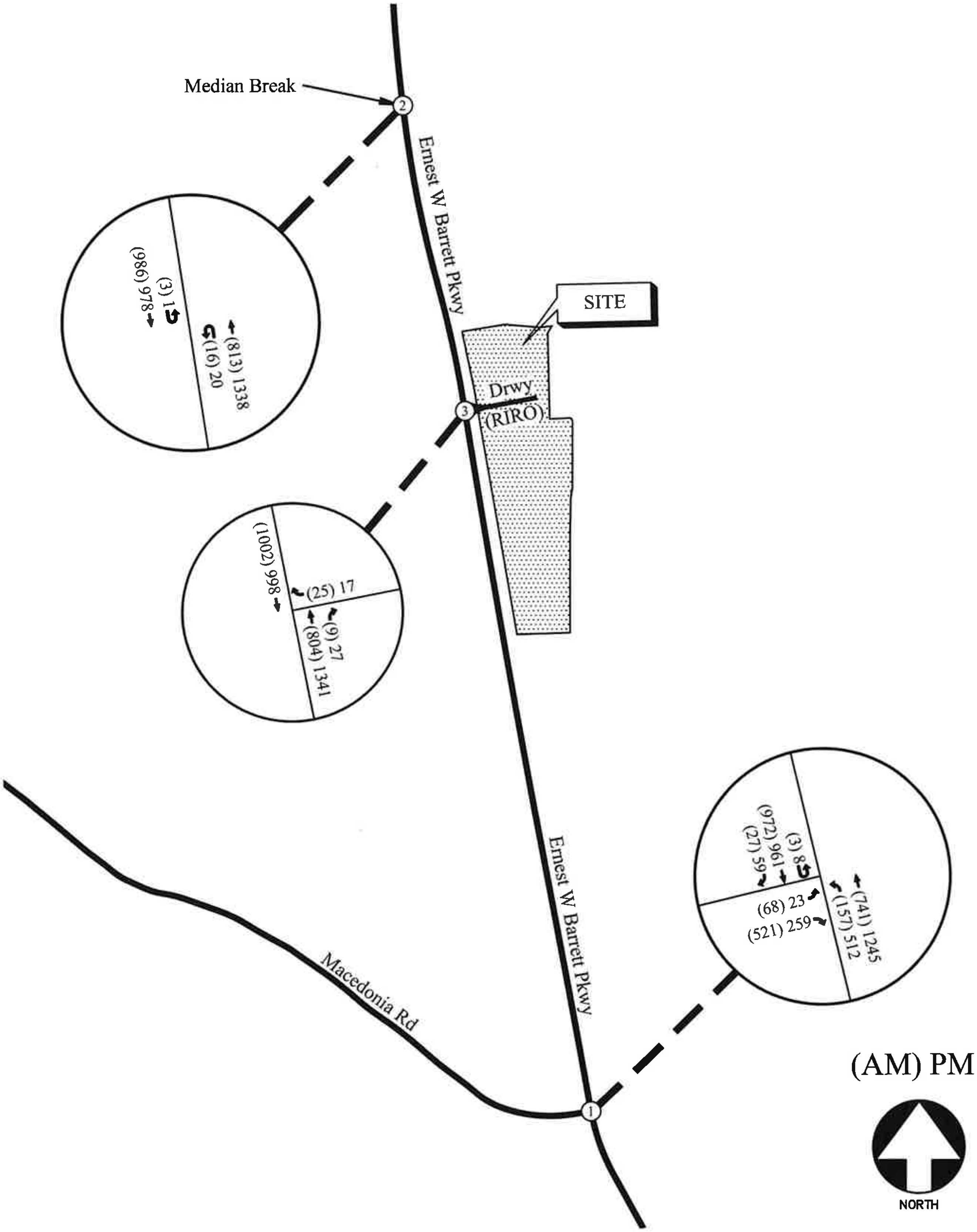
The "Build" or development conditions include the estimated background traffic from the "No-Build" conditions plus the added traffic from the proposed development. To evaluate future traffic operations in this area, the additional traffic volumes from the site (Figure 5) were added to base traffic volumes (Figure 6) to calculate the future traffic volumes after the construction of the development. These total future "Build" traffic volumes are shown in Figure 7.



FUTURE (NO-BUILD) WEEKDAY PEAK HOUR VOLUMES

FIGURE 6

A&R Engineering Inc.



FUTURE (BUILD) WEEKDAY PEAK HOUR VOLUMES FIGURE 7

A&R Engineering Inc.

6.3 Auxiliary Lane Analysis

Included below is the analysis for a right turn lane for the site driveway (per Cobb County Standards). The analyses below are based off the trip distribution included in Section 5.2. According to the trip distribution, the 24-hour two-way volume for traffic entering and exiting the site are 612 vehicles.

6.3.1 Deceleration Turn Lane Analysis

Per Cobb County Development Standards, (402.9), right turn lanes are required on any roadway that is included in the county's major thoroughfare plan network. Since Ernest W Barrett Parkway is an arterial roadway and is included in Cobb County's Major Thoroughfare Plan, a right turn lane is required at site driveway.

6.4 Future Traffic Operations

The future "No-Build" and "Build" traffic operations were analysed using the volumes in Figures 6 and 7, respectively. The results of the future traffic operations analysis are shown below in Table 5. Recommendations on traffic control and lane geometry are shown in Figure 8.

TABLE 5 – FUTURE INTERSECTION OPERATIONS

	Intersection	LOS (Delay)			
		NO BUILD		BUILD	
		AM Peak	PM Peak	AM Peak	PM Peak
1	**Ernest W Barrett Parkway @ Macedonia Road**	**C (23.2)**	**C (22.4)**	**C (23.3)**	**C (22.7)**
	-Eastbound Approach	D (54.2)	E (74.6)	D (54.3)	E (74.8)
	-Northbound Approach	A (9.2)	B (16.5)	A (9.2)	B (16.8)
	-Southbound Approach	B (17.3)	B (17.9)	B (17.5)	B (18.5)
2	**Ernest W Barrett Parkway @ Median Break**				
	-Northbound U-Turn	A (0.0)	A (0.0)	A (0.0)	A (0.0)
	-Southbound U-Turn	A (0.0)	A (0.0)	A (0.0)	A (0.0)
3	**Ernest W Barrett Parkway @ Site Driveway (Right In/Right Out)**				
	-Westbound Approach	-	-	B (11.7)	C (15.4)

The results of the future "No-Build" and "Build" traffic operations analysis indicate that the unsignalized study intersections will continue to operate at a level of service "C" or better in both the AM and PM peak hours. The signalized study intersection will be operating at a level of service "C" or better in both the AM and PM peak hour.

Table 6 below includes 95th percentile SimTraffic queue lengths at the study intersections. SimTraffic results of the "No Build" and "Build" have been used since the microsimulation more accurately modelled a multi-lane turn lane scenario than Synchro (macrosimulation).

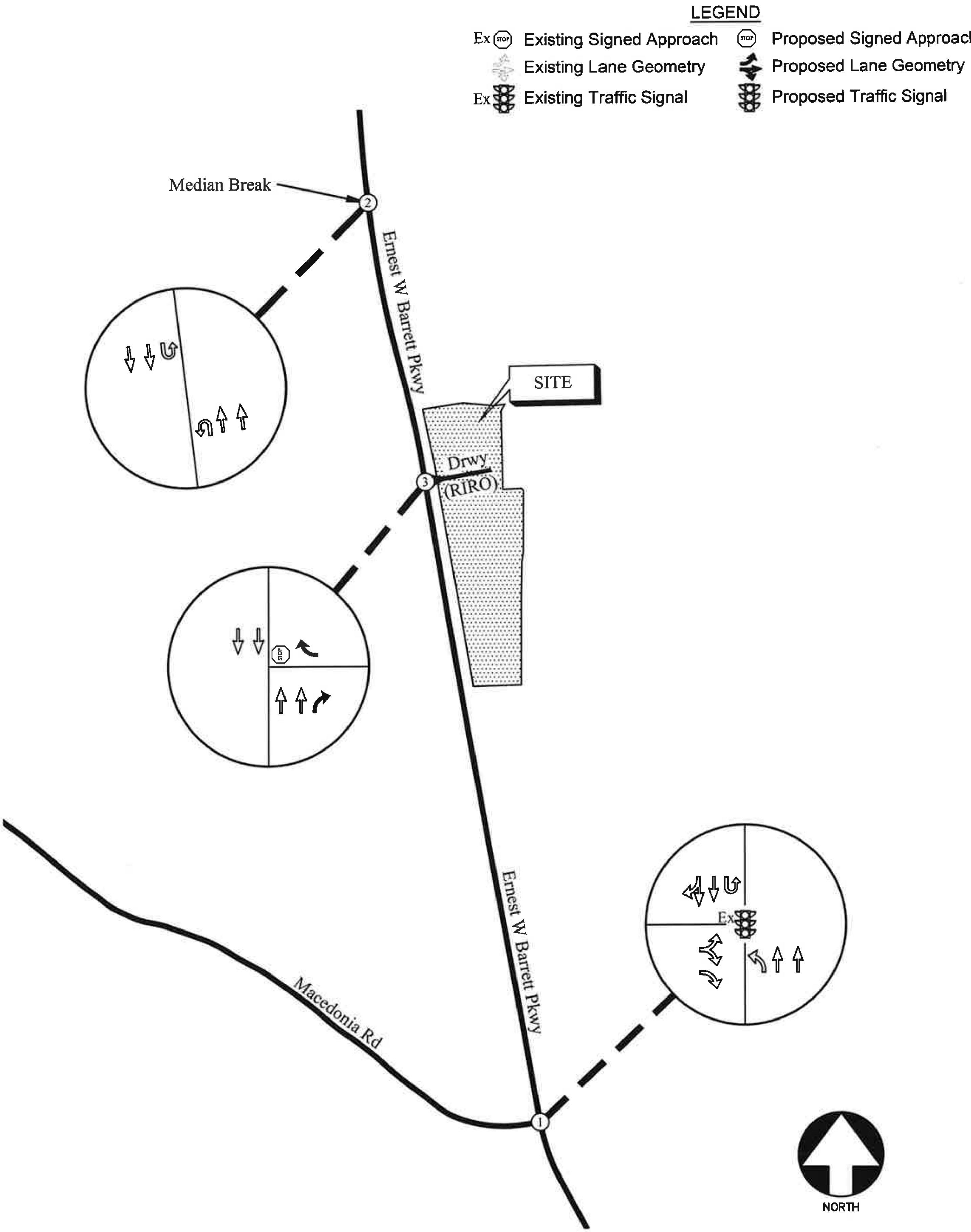
TABLE 6 – FUTURE INTERSECTION 95TH PERCENTILE QUEUES

Intersection		Available Storage	Queue Length in feet			
			NO-BUILD		BUILD	
			AM Peak	PM Peak	AM Peak	PM Peak
1	**Ernest W Barrett Parkway @ Macedonia Road**					
	-Eastbound Left/Right	460'	227	94	229	97
	-Eastbound Right	-	76	73	76	73
	-Northbound Left	305'	77	454	78	464
	-Northbound Through	-	172	183	174	190
	-Southbound U-Turn	190'	-	-	7	16
	-Southbound Through	-	411	504	419	511
	-Southbound Through/ Right	-	411	504	419	511
2	**Ernest W Barrett Parkway @ Median Break**					
	-Northbound U-Turn	145'	4	23	35	32
	-Southbound U-Turn	155'	11	6	12	5

The storage lengths are all adequate to accommodate projected queues other than the northbound left approach at the intersection of Ernest W Barrett Parkway and Macedonia Road during the PM peak hour.

6.5 System Improvements

The "No-Build" scenario shows a significant amount of northbound left turning volume at Ernest W Barrett Parkway and Macedonia Road during the PM peak hour. The expected 95th percentile queue during this time is also longer than the available storage for this movement. It is recommended to consider a dual left turn lane whenever the left turn volume is greater than 300 vehicles during the peak hour. The PM peak hour volume for northbound left turning vehicles at this intersection is 512.



FUTURE TRAFFIC CONTROL AND LANE GEOMETRY

FIGURE 8

A&R Engineering Inc.

7.0 Conclusions and Recommendations

Traffic impacts were evaluated for the proposed Somerset at Macedonia residential development that will be located on the east side of Ernest W Barrett Parkway in Cobb County. The development will consist of 43 single-family detached houses and proposes one right-in/right-out driveway on Ernest W Barrett Parkway.

Existing and future operations after completion of the project were analyzed at the intersections of:

1. Ernest W Barrett Parkway at Macedonia Road
2. Ernest W Barrett Parkway at Median Break (Approximately 0.45 miles to the north of the proposed site driveway location)
3. Ernest W Barrett Parkway at Site Driveway (Right In/Right Out)

The analysis included the evaluation of future operations for "No-Build" and "Build" conditions, with the differences between "No-Build" and "Build" accounting for an increase in traffic due to the proposed development. The results of the future "No-Build" and "Build" traffic operations analysis indicate that the unsignalized study intersections will continue to operate at a level of service "C" in both the AM and PM peak hours. The results of the future "No-Build" and "Build" traffic operations analysis also indicate that the signalized study intersections will continue to operate at a level of service "C" in both the AM and PM peak hours. The impact of the proposed site's traffic on the traffic operations in the study network is minimal.

7.1 Recommendation for Site Access Configuration

The following access configuration is recommended for the proposed site driveway intersection.

- Site Driveway: Right-In/ Right-Out driveway on Ernest W Barrett Parkway
  - One entering and one exiting lanes.
  - Stop-sign controlled on the driveway approach with Ernest W Barrett Parkway remaining free flow.
  - Deceleration turn lane for entering traffic
  - Provide adequate sight distance per AASHTO standards.

Appendix

Existing Intersection Traffic Counts

A & R Engineering, Inc.

2160 Kingston Court Suite 'O'
Marietta, GA 30067

TMC Data
Ernest W Barrett Pkwy @ Median Opening
North of Macedonia Road
7-9 am | 4-6 pm

File Name : 20250374
Site Code : 20250374
Start Date : 10-22-2025
Page No : 1

Groups Printed- Cars, Buses & Trucks

	Ernest W Barrett Pkwy Northbound					Ernest W Barrett Pkwy Southbound					Eastbound				Westbound				
Start Time	Left	Thru	Right	U-Turn	App. Total	Left	Thru	Right	U-Turn	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Int. Total
07:00 AM	0	191	0	0	191	0	208	0	2	210	0	0	0	0	0	0	0	0	401
07:15 AM	0	209	0	0	209	0	258	0	0	258	0	0	0	0	0	0	0	0	467
07:30 AM	0	204	0	0	204	0	265	0	0	265	0	0	0	0	0	0	0	0	469
07:45 AM	0	183	0	1	184	0	233	0	1	234	0	0	0	0	0	0	0	0	418
Total	0	787	0	1	788	0	964	0	3	967	0	0	0	0	0	0	0	0	1755
08:00 AM	0	169	0	1	170	0	228	0	1	229	0	0	0	0	0	0	0	0	399
08:15 AM	0	168	0	2	170	0	232	0	0	232	0	0	0	0	0	0	0	0	402
08:30 AM	0	147	0	1	148	0	221	0	2	223	0	0	0	0	0	0	0	0	371
08:45 AM	0	172	0	0	172	0	263	0	0	263	0	0	0	0	0	0	0	0	435
Total	0	656	0	4	660	0	944	0	3	947	0	0	0	0	0	0	0	0	1607
*** BREAK ***																			
04:00 PM	0	308	0	2	310	0	232	0	0	232	0	0	0	0	0	0	0	0	542
04:15 PM	0	321	0	3	324	0	216	0	0	216	0	0	0	0	0	0	0	0	540
04:30 PM	0	323	0	2	325	0	230	0	0	230	0	0	0	0	0	0	0	0	555
04:45 PM	0	354	0	3	357	0	248	0	1	249	0	0	0	0	0	0	0	0	606
Total	0	1306	0	10	1316	0	926	0	1	927	0	0	0	0	0	0	0	0	2243
05:00 PM	0	307	0	2	309	0	257	0	0	257	0	0	0	0	0	0	0	0	566
05:15 PM	0	290	0	2	292	0	222	0	1	223	0	0	0	0	0	0	0	0	515
05:30 PM	0	291	0	2	293	0	267	0	2	269	0	0	0	0	0	0	0	0	562
05:45 PM	0	280	0	2	282	0	240	0	1	241	0	0	0	0	0	0	0	0	523
Total	0	1168	0	8	1176	0	986	0	4	990	0	0	0	0	0	0	0	0	2166
Grand Total	0	3917	0	23	3940	0	3820	0	11	3831	0	0	0	0	0	0	0	0	7771
Apprch %	0	99.4	0	0.6		0	99.7	0	0.3		0	0	0		0	0	0		
Total %	0	50.4	0	0.3	50.7	0	49.2	0	0.1	49.3	0	0	0	0	0	0	0	0	

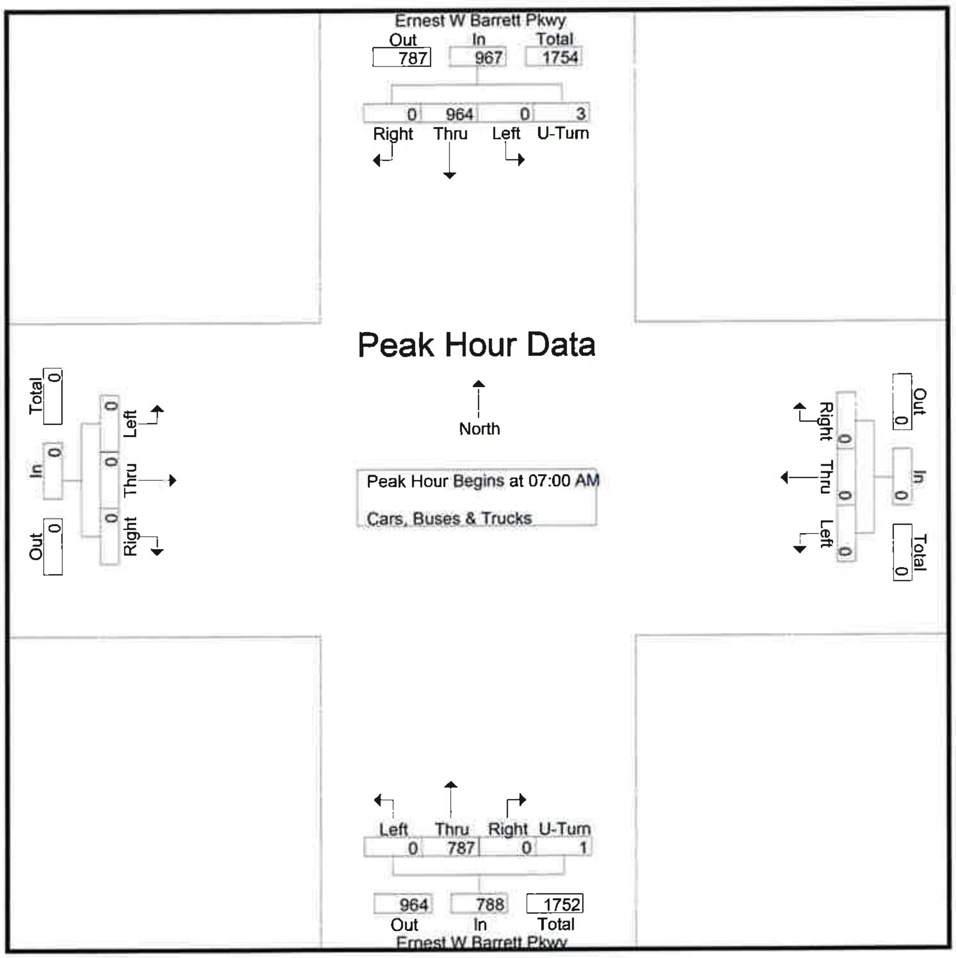
A & R Engineering, Inc.

2160 Kingston Court Suite 'O'
Marietta, GA 30067

TMC Data
Ernest W Barrett Pkwy @ Median Opening
North of Macedonia Road
7-9 am | 4-6 pm

File Name : 20250374
Site Code : 20250374
Start Date : 10-22-2025

	Ernest W Barrett Pkwy Northbound					Ernest W Barrett Pkwy Southbound					Eastbound				Westbound				
Start Time	Left	Thru	Right	U-Turn	App. Total	Left	Thru	Right	U-Turn	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Int. Total
Peak Hour Analysis From 07:00 AM to 08:45 AM - Peak 1 of 1																			
Peak Hour for Entire Intersection Begins at 07:00 AM																			
07:00 AM	0	191	0	0	191	0	208	0	**2**	210	0	0	0	0	0	0	0	0	401
07:15 AM	0	**209**	0	0	**209**	0	258	0	0	258	0	0	0	0	0	0	0	0	467
07:30 AM	0	204	0	0	204	0	**265**	0	0	**265**	0	0	0	0	0	0	0	0	**469**
07:45 AM	0	183	0	**1**	184	0	233	0	1	234	0	0	0	0	0	0	0	0	418
Total Volume	0	787	0	1	788	0	964	0	3	967	0	0	0	0	0	0	0	0	1755
% App. Total	0	99.9	0	0.1		0	99.7	0	0.3		0	0	0		0	0	0		
PHF	.000	.941	.000	.250	.943	.000	.909	.000	.375	.912	.000	.000	.000	.000	.000	.000	.000	.000	.936



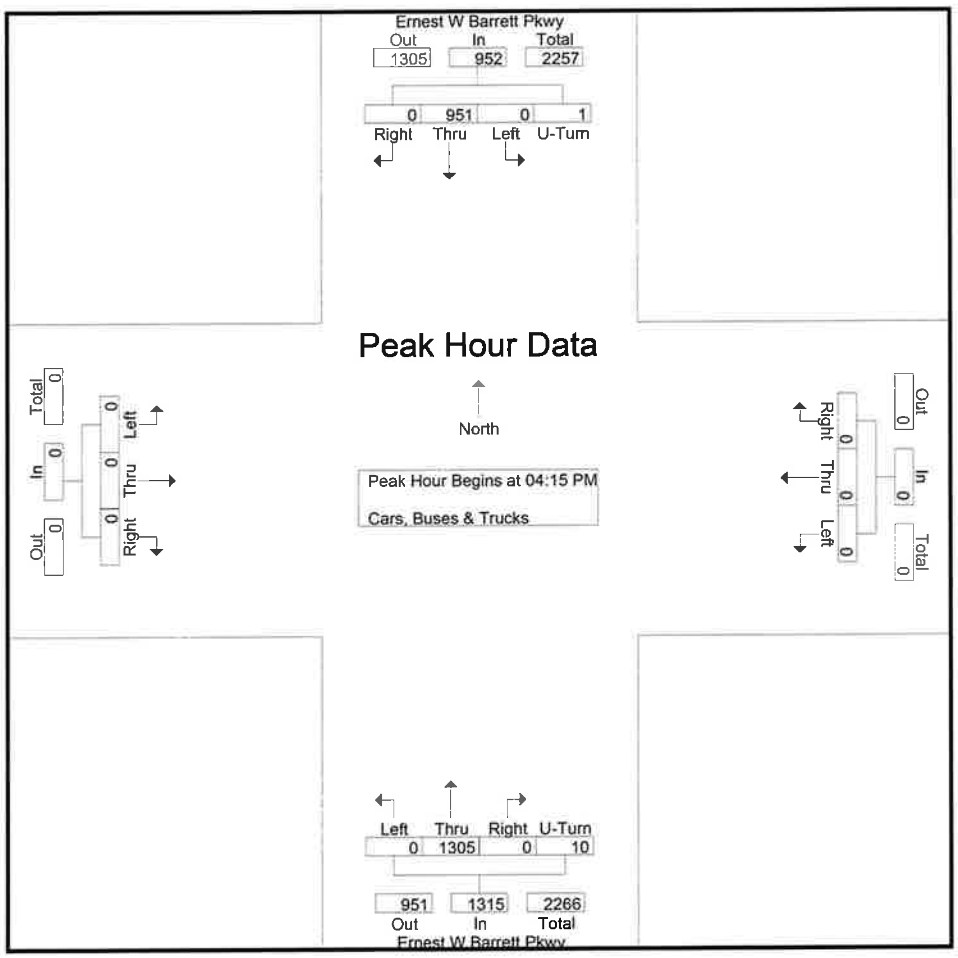
A & R Engineering, Inc.

2160 Kingston Court Suite 'O'
Marietta, GA 30067

TMC Data
Ernest W Barrett Pkwy @ Median Opening
North of Macedonia Road
7-9 am | 4-6 pm

File Name : 20250374
Site Code : 20250374
Start Date : 10-22-2025
Page No : 3

	Ernest W Barrett Pkwy Northbound					Ernest W Barrett Pkwy Southbound					Eastbound				Westbound				
Start Time	Left	Thru	Right	U-Turn	App. Total	Left	Thru	Right	U-Turn	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Int. Total
Peak Hour Analysis From 04:00 PM to 05:45 PM - Peak 1 of 1																			
Peak Hour for Entire Intersection Begins at 04:15 PM																			
04:15 PM	0	321	0	**3**	324	0	216	0	0	216	0	0	0	0	0	0	0	0	540
04:30 PM	0	323	0	2	325	0	230	0	0	230	0	0	0	0	0	0	0	0	555
04:45 PM	0	**354**	0	3	**357**	0	248	0	**1**	249	0	0	0	0	0	0	0	0	**606**
05:00 PM	0	307	0	2	309	0	**257**	0	0	**257**	0	0	0	0	0	0	0	0	566
Total Volume	0	1305	0	10	1315	0	951	0	1	952	0	0	0	0	0	0	0	0	2267
% App. Total	0	99.2	0	0.8		0	99.9	0	0.1		0	0	0		0	0	0		
PHF	.000	.922	.000	.833	.921	.000	.925	.000	.250	.926	.000	.000	.000	.000	.000	.000	.000	.000	.935



A & R Engineering, Inc.

2160 Kingston Court Suite 'O'
Marietta, GA 30067

TMC Data
Ernest W Barrett Pkwy @ Macedonia Road
7-9 am | 4-6 pm

File Name : 20250375
Site Code : 20250375
Start Date : 10-22-2025
Page No : 1

Groups Printed- Cars, Buses & Trucks

	Ernest W Barrett Pkwy Northbound				Ernest W Barrett Pkwy Southbound				Macedonia Road Eastbound				Westbound				
Start Time	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Int. Total
07:00 AM	12	165	0	177	0	196	12	208	26	0	151	177	0	0	0	0	562
07:15 AM	34	195	0	229	0	252	6	258	14	0	161	175	0	0	0	0	662
07:30 AM	45	190	0	235	0	265	0	265	14	0	111	125	0	0	0	0	625
07:45 AM	63	172	0	235	0	227	6	233	12	0	88	100	0	0	0	0	568
Total	154	722	0	876	0	940	24	964	66	0	511	577	0	0	0	0	2417
08:00 AM	40	160	0	200	0	223	5	228	10	0	91	101	0	0	0	0	529
08:15 AM	46	163	0	209	0	222	10	232	7	0	86	93	0	0	0	0	534
08:30 AM	32	141	0	173	0	217	4	221	7	0	72	79	0	0	0	0	473
08:45 AM	35	159	0	194	0	259	4	263	13	0	72	85	0	0	0	0	542
Total	153	623	0	776	0	921	23	944	37	0	321	358	0	0	0	0	2078
*** BREAK ***																	
04:00 PM	91	302	0	393	0	222	10	232	8	0	63	71	0	0	0	0	696
04:15 PM	88	317	0	405	0	205	11	216	7	0	57	64	0	0	0	0	685
04:30 PM	98	317	0	415	0	222	8	230	8	0	57	65	0	0	0	0	710
04:45 PM	101	353	0	454	0	233	15	248	4	0	53	57	0	0	0	0	759
Total	378	1289	0	1667	0	882	44	926	27	0	230	257	0	0	0	0	2850
05:00 PM	107	307	0	414	0	243	14	257	2	0	62	64	0	0	0	0	735
05:15 PM	130	287	0	417	0	212	10	222	5	0	63	68	0	0	0	0	707
05:30 PM	116	290	0	406	0	248	19	267	3	0	55	58	0	0	0	0	731
05:45 PM	149	321	0	470	0	230	13	243	10	0	74	84	0	0	0	0	797
Total	502	1205	0	1707	0	933	56	989	20	0	254	274	0	0	0	0	2970
Grand Total	1187	3839	0	5026	0	3676	147	3823	150	0	1316	1466	0	0	0	0	10315
Apprch %	23.6	76.4	0		0	96.2	3.8		10.2	0	89.8		0	0	0		
Total %	11.5	37.2	0	48.7	0	35.6	1.4	37.1	1.5	0	12.8	14.2	0	0	0	0	

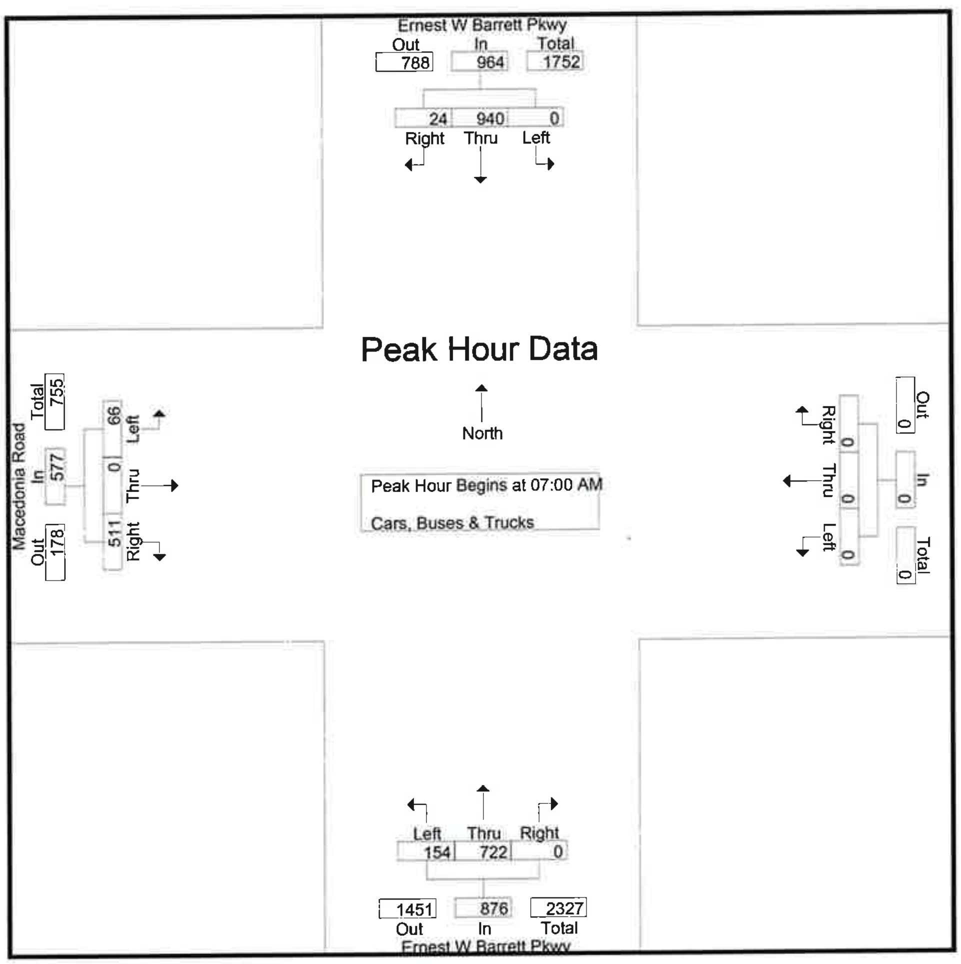
A & R Engineering, Inc.

2160 Kingston Court Suite 'O'
Marietta, GA 30067

TMC Data
Ernest W Barrett Pkwy @ Macedonia Road
7-9 am | 4-6 pm

File Name : 20250375
Site Code : 20250375
Start Date : 10-22-2025

	Ernest W Barrett Pkwy Northbound				Ernest W Barrett Pkwy Southbound				Macedonia Road Eastbound				Westbound				
Start Time	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Int. Total
Peak Hour Analysis From 07:00 AM to 08:45 AM - Peak 1 of 1																	
Peak Hour for Entire Intersection Begins at 07:00 AM																	
07:00 AM	12	165	0	177	0	196	**12**	208	**26**	0	151	**177**	0	0	0	0	562
07:15 AM	34	**195**	0	229	0	252	6	258	14	0	**161**	175	0	0	0	0	**662**
07:30 AM	45	190	0	**235**	0	**265**	0	**265**	14	0	111	125	0	0	0	0	625
07:45 AM	**63**	172	0	235	0	227	6	233	12	0	88	100	0	0	0	0	568
Total Volume	154	722	0	876	0	940	24	964	66	0	511	577	0	0	0	0	2417
% App. Total	17.6	82.4	0		0	97.5	2.5		11.4	0	88.6		0	0	0		
PHF	.611	.926	.000	.932	.000	.887	.500	.909	.635	.000	.793	.815	.000	.000	.000	.000	.913



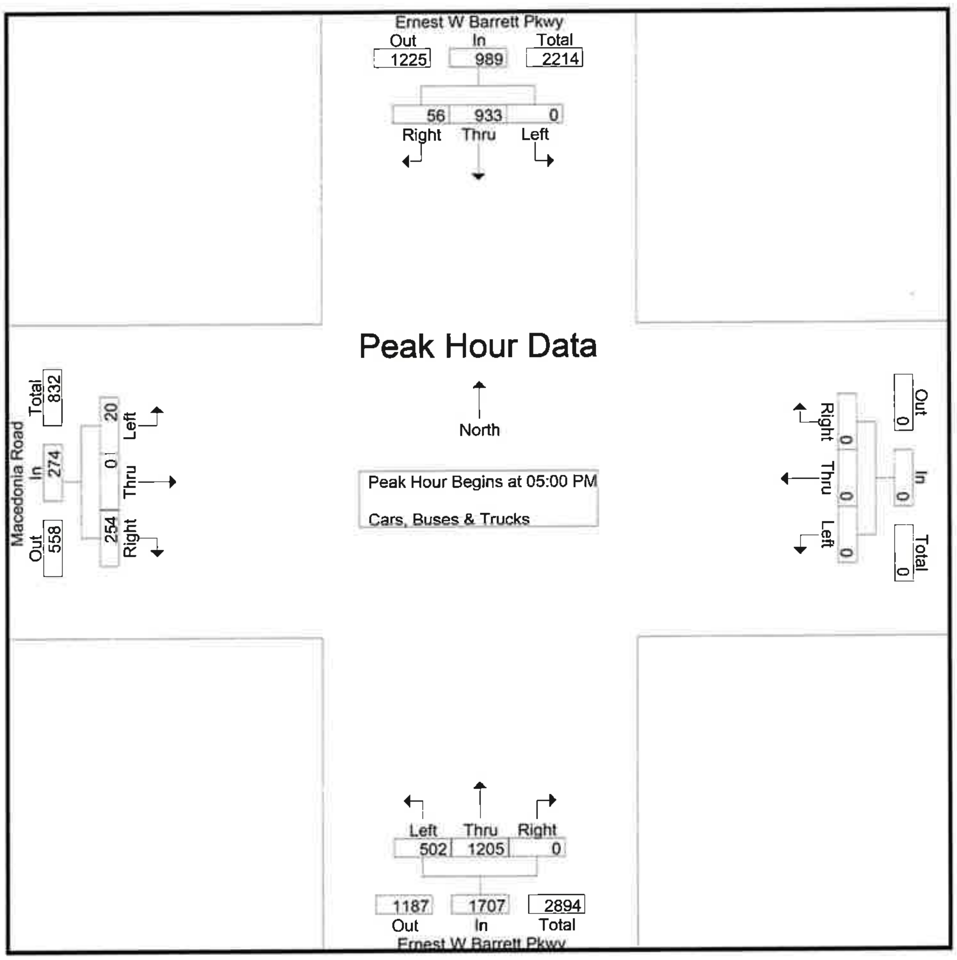
A & R Engineering, Inc.

2160 Kingston Court Suite 'O'
Marietta, GA 30067

TMC Data
Ernest W Barrett Pkwy @ Macedonia Road
7-9 am | 4-6 pm

File Name : 20250375
Site Code : 20250375
Start Date : 10-22-2025
Page No : 3

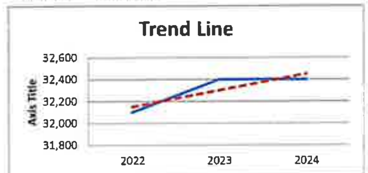
	Ernest W Barrett Pkwy Northbound				Ernest W Barrett Pkwy Southbound				Macedonia Road Eastbound				Westbound				
Start Time	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Left	Thru	Right	App. Total	Int. Total
Peak Hour Analysis From 04:00 PM to 05:45 PM - Peak 1 of 1																	
Peak Hour for Entire Intersection Begins at 05:00 PM																	
05:00 PM	107	307	0	414	0	243	14	257	2	0	62	64	0	0	0	0	735
05:15 PM	130	287	0	417	0	212	10	222	5	0	63	68	0	0	0	0	707
05:30 PM	116	290	0	406	0	**248**	**19**	**267**	3	0	55	58	0	0	0	0	731
05:45 PM	**149**	**321**	0	**470**	0	230	13	243	**10**	0	**74**	**84**	0	0	0	0	**797**
Total Volume	502	1205	0	1707	0	933	56	989	20	0	254	274	0	0	0	0	2970
% App. Total	29.4	70.6	0		0	94.3	5.7		7.3	0	92.7		0	0	0		
PHF	.842	.938	.000	.908	.000	.941	.737	.926	.500	.000	.858	.815	.000	.000	.000	.000	.932



Linear Regression of Daily Traffic

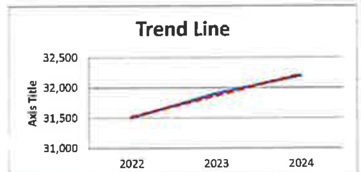
Location	Growth Rate	R Squared	Station ID	Route	2022	2023	2024
SR 360 (Macland Road) (W of W. Sandtown I	0.5%	0.75	067-0669	00000360	32,100	32,400	32,400
SR 360 (Macland Road) (W of Scotney Castle	1.1%	0.99	067-0667	00036000	31,500	31,900	32,200
Macedonia Road (SW of Hopkins Road)	2.0%	0.96	067-8836	0000000	5,440	5,510	5,660
East-West Connector (SE of Tramore Pointe I	2.8%	0.89	067-0770	00283500	37,700	38,100	39,800
Ernest W. Barrett Parkway (NW of Macedoni	-2.4%	0.62	067-0657	00822800	22,600	22,800	21,500
Weighted Average	**0.9%**	0.98	Sum of Count Stations =		1,29,340	1,30,710	1,31,560

Location	Traffic Counter	RCLINK	2022	2023	2024
SR 360 (Macland Road) (W of W. Sandtown Road)	067-0669	00000360	32,100	32,400	32,400



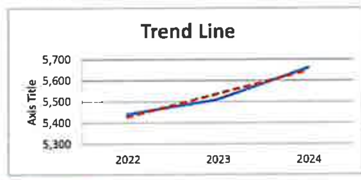
Growth Rate	0.5%	**Intercept**	-2,71,150	**Slope**	150.00
Trend Line			32,150	32,300	32,450

Location	Traffic Counter	RCLINK	2022	2023	2024
SR 360 (Macland Road) (W of Scotney Castle Lane)	067-0667	00036000	31,500	31,900	32,200



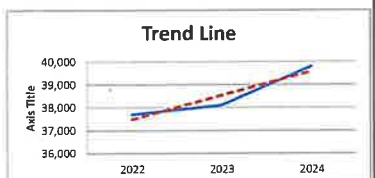
Growth Rate	1.1%	**Intercept**	-6,76,183	**Slope**	350.00
Trend Line			31,517	31,867	32,217

Location	Traffic Counter	RCLINK	2022	2023	2024
Macedonia Road (SW of Hopkins Road)	067-8836	0000000	5,440	5,510	5,660



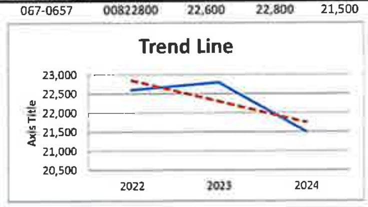
Growth Rate	2.0%	**Intercept**	-2,16,993	**Slope**	110.00
Trend Line			5,427	5,537	5,647

Location	Traffic Counter	RCLINK	2022	2023	2024
East-West Connector (SE of Tramore Pointe Parkway/ Asquith Avenue)	067-0770	00283500	37,700	38,100	39,800



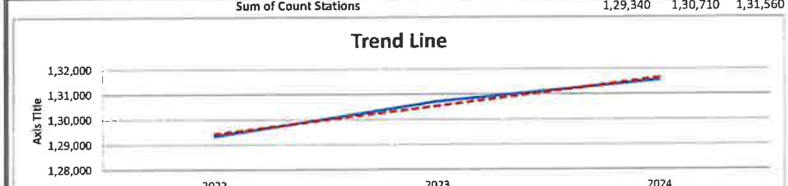
Growth Rate	2.8%	**Intercept**	-20,85,617	**Slope**	1,050.00
Trend Line			37,483	38,533	39,583

Location	Traffic Counter	RCLINK	2022	2023	2024
Ernest W. Barrett Parkway (NW of Macedonia Road)	067-0657	00822800	22,600	22,800	21,500



Growth Rate	-2.4%	**Intercept**	11,34,950	**Slope**	-550.00
Trend Line			22,850	22,300	21,750

	2022	2023	2024
Weighted Average			
Sum of Count Stations	1,29,340	1,30,710	1,31,560



Growth Rate	0.9%	**Intercept**	-21,14,993	**Slope**	1,110.00
Trend Line			1,29,427	1,30,537	1,31,647

Existing Intersection Analysis

Lane Group	EBL	EBR	NBL	NBT	SBT
Lane Configurations					
Traffic Volume (vph)	66	511	154	722	940
Future Volume (vph)	66	511	154	722	940
Lane Group Flow (vph)	320	315	169	793	1059
Turn Type	Prot	Prot	pm+pt	NA	NA
Protected Phases	8	8	1	6	2
Permitted Phases			6		
Detector Phase	8	8	1	6	2
Switch Phase					
Minimum Initial (s)	6.0	6.0	5.0	15.0	15.0
Minimum Split (s)	23.5	23.5	15.0	23.5	40.5
Total Split (s)	40.0	40.0	30.0	80.0	50.0
Total Split (%)	33.3%	33.3%	25.0%	66.7%	41.7%
Yellow Time (s)	3.5	3.5	3.5	3.5	3.5
All-Red Time (s)	2.0	2.0	2.0	2.0	2.0
Lost Time Adjust (s)	0.0	0.0	0.0	0.0	0.0
Total Lost Time (s)	5.5	5.5	5.5	5.5	5.5
Lead/Lag			Lead		Lag
Lead-Lag Optimize?			Yes		Yes
Recall Mode	None	None	None	C-Min	C-Min
v/c Ratio	0.83	0.62	0.42	0.30	0.50
Control Delay (s/veh)	43.4	9.7	8.7	6.3	16.5
Queue Delay	0.0	0.0	0.0	0.0	0.0
Total Delay (s/veh)	43.4	9.7	8.7	6.3	16.5
Queue Length 50th (ft)	138	0	32	92	228
Queue Length 95th (ft)	222	76	74	166	392
Internal Link Dist (ft)	433			445	3140
Turn Bay Length (ft)	135		315		
Base Capacity (vph)	569	656	558	2629	2122
Starvation Cap Reductn	0	0	0	0	0
Spillback Cap Reductn	0	0	0	0	0
Storage Cap Reductn	0	0	0	0	0
Reduced v/c Ratio	0.56	0.48	0.30	0.30	0.50

Intersection Summary

Cycle Length: 120
Actuated Cycle Length: 120
Offset: 0 (0%), Referenced to phase 2:SBTU and 6:NBTL, Start of Green
Natural Cycle: 80
Control Type: Actuated-Coordinated

Splits and Phases: 1: Ernest W. Barrett Pkwy & Macedonia Rd

Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	66	511	154	722	0	940	24
Future Volume (veh/h)	66	511	154	722	0	940	24
Initial Q (Qb), veh	0	0	0	0		0	0
Ped-Bike Adj(A_pbT)	1.00	1.00	1.00				1.00
Parking Bus, Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	640	169	793		1033	26
Peak Hour Factor	0.91	0.91	0.91	0.91		0.91	0.91
Percent Heavy Veh, %	2	2	2	2		2	2
Cap, veh/h	404	719	376	2422		2055	52
Arrive On Green	0.00	0.23	0.06	0.68		0.58	0.58
Sat Flow, veh/h	1781	3170	1781	3647		3636	89
Grp Volume(v), veh/h	0	640	169	793		518	541
Grp Sat Flow(s),veh/h/ln	1781	1585	1781	1777		1777	1854
Q Serve(g_s), s	0.0	23.5	4.4	11.0		20.7	20.7
Cycle Q Clear(g_c), s	0.0	23.5	4.4	11.0		20.7	20.7
Prop In Lane	1.00	1.00	1.00				0.05
Lane Grp Cap(c), veh/h	404	719	376	2422		1031	1076
V/C Ratio(X)	0.00	0.89	0.45	0.33		0.50	0.50
Avail Cap(c_a), veh/h	512	911	641	2422		1031	1076
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Upstream Filter(I)	0.00	1.00	1.00	1.00		1.00	1.00
Uniform Delay (d), s/veh	0.0	45.0	11.2	7.8		14.9	14.9
Incr Delay (d2), s/veh	0.0	9.1	0.8	0.4		1.8	1.7
Initial Q Delay(d3), s/veh	0.0	0.0	0.0	0.0		0.0	0.0
%ile BackOfQ(50%),veh/ln	0.0	10.0	1.6	3.7		8.1	8.4
Unsig. Movement Delay, s/veh							
LnGrp Delay(d), s/veh	0.0	54.1	12.0	8.2		16.7	16.6
LnGrp LOS		D	B	A		B	B
Approach Vol, veh/h	640			962		1059	
Approach Delay, s/veh	54.1			8.9		16.7	
Approach LOS	D			A		B	

Timer - Assigned Phs	1	2	6	8
Phs Duration (G+Y+Rc), s	12.2	75.1	87.3	32.7
Change Period (Y+Rc), s	5.5	5.5	5.5	5.5
Max Green Setting (Gmax), s	24.5	44.5	74.5	34.5
Max Q Clear Time (g_c+I1), s	6.4	22.7	13.0	25.5
Green Ext Time (p_c), s	0.3	9.0	8.7	1.7

Intersection Summary	
HCM 6th Ctrl Delay, s/veh	22.8
HCM 6th LOS	C

Notes

User approved volume balancing among the lanes for turning movement.
User approved ignoring U-Turning movement.

Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	66	511	154	722	0	940	24
Future Volume (veh/h)	66	511	154	722	0	940	24
Number	3	18	1	6		2	12
Initial Q, veh	0	0	0	0		0	0
Ped-Bike Adj (A_pbT)	1.00	1.00	1.00				1.00
Parking Bus Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Lanes Open During Work Zone							
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	640	169	793		1033	26
Peak Hour Factor	0.91	0.91	0.91	0.91		0.91	0.91
Percent Heavy Veh, %	2	2	2	2		2	2
Opposing Right Turn Influence	Yes		Yes				
Cap, veh/h	404	719	376	2422		2055	52
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Prop Arrive On Green	0.00	0.23	0.06	0.68		0.58	0.58
Unsig. Movement Delay							
Ln Grp Delay, s/veh	0.0	54.1	12.0	8.2		16.7	16.6
Ln Grp LOS		D	B	A		B	B
Approach Vol, veh/h	640			962		1059	
Approach Delay, s/veh	54.1			8.9		16.7	
Approach LOS	D			A		B	

Timer:	1	2	3	4	5	6	7	8
Assigned Phs	1	2	8			6		
Case No	1.2	8.0	9.0			4.0		
Phs Duration (G+Y+Rc), s	12.2	75.1	32.7			87.3		
Change Period (Y+Rc), s	5.5	5.5	5.5			5.5		
Max Green (Gmax), s	24.5	44.5	34.5			74.5		
Max Allow Headway (MAH), s	3.5	5.8	3.8			5.8		
Max Q Clear (g_c+I1), s	6.4	22.7	25.5			13.0		
Green Ext Time (g_e), s	0.3	9.0	1.7			8.7		
Prob of Phs Call (p_c)	1.00	1.00	1.00			1.00		
Prob of Max Out (p_x)	0.00	0.30	0.12			0.00		
Left-Turn Movement Data								
Assigned Mvmt	1	5	3					
Mvmt Sat Flow, veh/h	1781	0	1781					
Through Movement Data								
Assigned Mvmt		2	8			6		
Mvmt Sat Flow, veh/h		3636	0			3647		
Right-Turn Movement Data								
Assigned Mvmt		12	18			16		
Mvmt Sat Flow, veh/h		89	3170			0		
Left Lane Group Data								
Assigned Mvmt	1	5	3	0	0	0	0	0
Lane Assignment	L (Pr/Pm)		L					

Lanes in Grp	1	0	1	0	0	0	0	0
Grp Vol (v), veh/h	169	0	0	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	1781	0	1781	0	0	0	0	0
Q Serve Time (g_s), s	4.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	4.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Sat Flow (s_l), veh/h/ln	533	0	1781	0	0	0	0	0
Shared LT Sat Flow (s_sh), veh/h/ln	0	0	0	0	0	0	0	0
Perm LT Eff Green (g_p), s	71.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Serve Time (g_u), s	48.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Q Serve Time (g_ps), s	10.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Time to First Blk (g_f), s	0.0	69.6	0.0	0.0	0.0	0.0	0.0	0.0
Serve Time pre Blk (g_fs), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop LT Inside Lane (P_L)	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	376	0	404	0	0	0	0	0
V/C Ratio (X)	0.45	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	641	0	512	0	0	0	0	0
Upstream Filter (I)	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	11.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	12.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
%ile Back of Q (50%), veh/ln	1.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Middle Lane Group Data								
Assigned Mvmt	0	2	8	0	0	6	0	0
Lane Assignment		T				T		
Lanes in Grp	0	1	0	0	0	2	0	0
Grp Vol (v), veh/h	0	518	0	0	0	793	0	0
Grp Sat Flow (s), veh/h/ln	0	1777	0	0	0	1777	0	0
Q Serve Time (g_s), s	0.0	20.7	0.0	0.0	0.0	11.0	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	20.7	0.0	0.0	0.0	11.0	0.0	0.0
Lane Grp Cap (c), veh/h	0	1031	0	0	0	2422	0	0
V/C Ratio (X)	0.00	0.50	0.00	0.00	0.00	0.33	0.00	0.00
Avail Cap (c_a), veh/h	0	1031	0	0	0	2422	0	0
Upstream Filter (I)	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	14.9	0.0	0.0	0.0	7.8	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.8	0.0	0.0	0.0	0.4	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	16.7	0.0	0.0	0.0	8.2	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	7.6	0.0	0.0	0.0	3.6	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.0	0.0	0.0	0.1	0.0	0.0

3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	8.1	0.0	0.0	0.0	3.7	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.07	0.00	0.00	0.00	0.19	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Right Lane Group Data								
Assigned Mvmt	0	12	18	0	0	16	0	0
Lane Assignment		T+R	R					
Lanes in Grp	0	1	2	0	0	0	0	0
Grp Vol (v), veh/h	0	541	640	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	0	1854	1585	0	0	0	0	0
Q Serve Time (g_s), s	0.0	20.7	23.5	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	20.7	23.5	0.0	0.0	0.0	0.0	0.0
Prot RT Sat Flow (s_R), veh/h/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prot RT Eff Green (g_R), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop RT Outside Lane (P_R)	0.00	0.05	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	0	1076	719	0	0	0	0	0
V/C Ratio (X)	0.00	0.50	0.89	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	0	1076	911	0	0	0	0	0
Upstream Filter (I)	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	14.9	45.0	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.7	9.1	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	16.6	54.1	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	7.9	9.1	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.9	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	8.4	10.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.07	0.55	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Intersection Summary

HCM 6th Ctrl Delay, s/veh	22.8
HCM 6th LOS	C

Notes

User approved volume balancing among the lanes for turning movement.
User approved ignoring U-Turning movement.

Lane Group	EBL	EBR	NBL	NBT	SBT
Lane Configurations					
Traffic Volume (vph)	20	254	502	1205	933
Future Volume (vph)	20	254	502	1205	933
Lane Group Flow (vph)	148	147	540	1296	1063
Turn Type	Prot	Prot	pm+pt	NA	NA
Protected Phases	8	8	1	6	2
Permitted Phases			6		
Detector Phase	8	8	1	6	2
Switch Phase					
Minimum Initial (s)	6.0	6.0	5.0	15.0	15.0
Minimum Split (s)	23.5	23.5	15.0	23.5	40.5
Total Split (s)	30.0	30.0	45.0	120.0	75.0
Total Split (%)	20.0%	20.0%	30.0%	80.0%	50.0%
Yellow Time (s)	3.5	3.5	3.5	3.5	3.5
All-Red Time (s)	2.0	2.0	2.0	2.0	2.0
Lost Time Adjust (s)	0.0	0.0	0.0	0.0	0.0
Total Lost Time (s)	5.5	5.5	5.5	5.5	5.5
Lead/Lag			Lead		Lag
Lead-Lag Optimize?			Yes		Yes
Recall Mode	None	None	None	C-Min	C-Min
v/c Ratio	0.69	0.64	0.78	0.42	0.56
Control Delay (s/veh)	31.8	22.5	29.0	2.8	25.3
Queue Delay	0.0	0.0	0.0	0.0	0.0
Total Delay (s/veh)	31.8	22.5	29.0	2.8	25.3
Queue Length 50th (ft)	21	0	256	94	350
Queue Length 95th (ft)	92	73	420	176	490
Internal Link Dist (ft)	433			445	3140
Turn Bay Length (ft)	135		315		
Base Capacity (vph)	368	368	707	3064	1905
Starvation Cap Reductn	0	0	0	0	0
Spillback Cap Reductn	0	0	0	0	0
Storage Cap Reductn	0	0	0	0	0
Reduced v/c Ratio	0.40	0.40	0.76	0.42	0.56

Intersection Summary
Cycle Length: 150
Actuated Cycle Length: 150
Offset: 0 (0%), Referenced to phase 2:SBTU and 6:NBTL, Start of Green
Natural Cycle: 90
Control Type: Actuated-Coordinated

Splits and Phases: 1: Ernest W. Barrett Pkwy & Macedonia Rd

Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR	
Lane Configurations								
Traffic Volume (veh/h)	20	254	502	1205	0	933	56	
Future Volume (veh/h)	20	254	502	1205	0	933	56	
Initial Q (Qb), veh	0	0	0	0		0	0	
Ped-Bike Adj(A_pbT)	1.00	1.00	1.00				1.00	
Parking Bus, Adj	1.00	1.00	1.00	1.00		1.00	1.00	
Work Zone On Approach	No			No		No		
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870	
Adj Flow Rate, veh/h	0	297	540	1296		1003	60	
Peak Hour Factor	0.93	0.93	0.93	0.93		0.93	0.93	
Percent Heavy Veh, %	2	2	2	2		2	2	
Cap, veh/h	196	348	568	2903		2145	128	
Arrive On Green	0.00	0.11	0.15	0.82		0.63	0.63	
Sat Flow, veh/h	1781	3170	1781	3647		3500	204	
Grp Volume(v), veh/h	0	297	540	1296		523	540	
Grp Sat Flow(s),veh/h/ln	1781	1585	1781	1777		1777	1834	
Q Serve(g_s), s	0.0	13.8	19.2	15.8		23.2	23.2	
Cycle Q Clear(g_c), s	0.0	13.8	19.2	15.8		23.2	23.2	
Prop In Lane	1.00	1.00	1.00				0.11	
Lane Grp Cap(c), veh/h	196	348	568	2903		1119	1155	
V/C Ratio(X)	0.00	0.85	0.95	0.45		0.47	0.47	
Avail Cap(c_a), veh/h	291	518	769	2903		1119	1155	
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00	
Upstream Filter(I)	0.00	1.00	1.00	1.00		1.00	1.00	
Uniform Delay (d), s/veh	0.0	65.6	23.1	4.0		14.6	14.6	
Incr Delay (d2), s/veh	0.0	8.8	17.9	0.5		1.4	1.4	
Initial Q Delay(d3), s/veh	0.0	0.0	0.0	0.0		0.0	0.0	
%ile BackOfQ(50%),veh/ln	0.0	6.0	22.1	4.3		9.2	9.5	
Unsig. Movement Delay, s/veh								
LnGrp Delay(d), s/veh	0.0	74.4	41.0	4.5		16.0	15.9	
LnGrp LOS		E	D	A		B	B	
Approach Vol, veh/h	297			1836		1063		
Approach Delay, s/veh	74.4			15.2		16.0		
Approach LOS	E			B		B		
Timer - Assigned Phs	1	2				6		8
Phs Duration (G+Y+Rc), s	28.1	100.0				128.0		22.0
Change Period (Y+Rc), s	5.5	5.5				5.5		5.5
Max Green Setting (Gmax), s	39.5	69.5				114.5		24.5
Max Q Clear Time (g_c+I1), s	21.2	25.2				17.8		15.8
Green Ext Time (p_c), s	1.4	11.6				20.0		0.7

Intersection Summary		
HCM 6th Ctrl Delay, s/veh		21.0
HCM 6th LOS		C

Notes

User approved pedestrian interval to be less than phase max green.
User approved volume balancing among the lanes for turning movement.
User approved ignoring U-Turning movement.

Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	20	254	502	1205	0	933	56
Future Volume (veh/h)	20	254	502	1205	0	933	56
Number	3	18	1	6		2	12
Initial Q, veh	0	0	0	0		0	0
Ped-Bike Adj (A_pbT)	1.00	1.00	1.00				1.00
Parking Bus Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Lanes Open During Work Zone							
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	297	540	1296		1003	60
Peak Hour Factor	0.93	0.93	0.93	0.93		0.93	0.93
Percent Heavy Veh, %	2	2	2	2		2	2
Opposing Right Turn Influence	Yes		Yes				
Cap, veh/h	196	348	568	2903		2145	128
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Prop Arrive On Green	0.00	0.11	0.15	0.82		0.63	0.63
Unsig. Movement Delay							
Ln Grp Delay, s/veh	0.0	74.4	41.0	4.5		16.0	15.9
Ln Grp LOS		E	D	A		B	B
Approach Vol, veh/h	297			1836		1063	
Approach Delay, s/veh	74.4			15.2		16.0	
Approach LOS	E			B		B	

Timer:	1	2	3	4	5	6	7	8
Assigned Phs	1	2	8			6		
Case No	1.2	8.0	9.0			4.0		
Phs Duration (G+Y+Rc), s	28.1	100.0	22.0			128.0		
Change Period (Y+Rc), s	5.5	5.5	5.5			5.5		
Max Green (Gmax), s	39.5	69.5	24.5			114.5		
Max Allow Headway (MAH), s	3.5	5.8	3.8			5.8		
Max Q Clear (g_c+I1), s	21.2	25.2	15.8			17.8		
Green Ext Time (g_e), s	1.4	11.6	0.7			20.0		
Prob of Phs Call (p_c)	1.00	1.00	1.00			1.00		
Prob of Max Out (p_x)	0.00	0.06	0.03			0.01		
Left-Turn Movement Data								
Assigned Mvmt	1	5	3					
Mvmt Sat Flow, veh/h	1781	0	1781					
Through Movement Data								
Assigned Mvmt		2	8			6		
Mvmt Sat Flow, veh/h		3500	0			3647		
Right-Turn Movement Data								
Assigned Mvmt		12	18			16		
Mvmt Sat Flow, veh/h		204	3170			0		
Left Lane Group Data								
Assigned Mvmt	1	5	3	0	0	0	0	0
Lane Assignment	L (Pr/Pm)		L					

Lanes in Grp	1	0	1	0	0	0	0	0
Grp Vol (v), veh/h	540	0	0	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	1781	0	1781	0	0	0	0	0
Q Serve Time (g_s), s	19.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	19.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Sat Flow (s_l), veh/h/ln	531	0	1781	0	0	0	0	0
Shared LT Sat Flow (s_sh), veh/h/ln	0	0	0	0	0	0	0	0
Perm LT Eff Green (g_p), s	96.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Serve Time (g_u), s	71.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Q Serve Time (g_ps), s	71.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Time to First Blk (g_f), s	0.0	94.5	0.0	0.0	0.0	0.0	0.0	0.0
Serve Time pre Blk (g_fs), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop LT Inside Lane (P_L)	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	568	0	196	0	0	0	0	0
V/C Ratio (X)	0.95	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	769	0	291	0	0	0	0	0
Upstream Filter (I)	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	23.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	17.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	41.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	19.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	2.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
%ile Back of Q (50%), veh/ln	22.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	1.78	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Middle Lane Group Data								
Assigned Mvmt	0	2	8	0	0	6	0	0
Lane Assignment		T				T		
Lanes in Grp	0	1	0	0	0	2	0	0
Grp Vol (v), veh/h	0	523	0	0	0	1296	0	0
Grp Sat Flow (s), veh/h/ln	0	1777	0	0	0	1777	0	0
Q Serve Time (g_s), s	0.0	23.2	0.0	0.0	0.0	15.8	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	23.2	0.0	0.0	0.0	15.8	0.0	0.0
Lane Grp Cap (c), veh/h	0	1119	0	0	0	2903	0	0
V/C Ratio (X)	0.00	0.47	0.00	0.00	0.00	0.45	0.00	0.00
Avail Cap (c_a), veh/h	0	1119	0	0	0	2903	0	0
Upstream Filter (I)	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	14.6	0.0	0.0	0.0	4.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.4	0.0	0.0	0.0	0.5	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	16.0	0.0	0.0	0.0	4.5	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	8.8	0.0	0.0	0.0	4.1	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.4	0.0	0.0	0.0	0.2	0.0	0.0

3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	9.2	0.0	0.0	0.0	4.3	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.07	0.00	0.00	0.00	0.23	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Right Lane Group Data								
Assigned Mvmt	0	12	18	0	0	16	0	0
Lane Assignment		T+R	R					
Lanes in Grp	0	1	2	0	0	0	0	0
Grp Vol (v), veh/h	0	540	297	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	0	1834	1585	0	0	0	0	0
Q Serve Time (g_s), s	0.0	23.2	13.8	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	23.2	13.8	0.0	0.0	0.0	0.0	0.0
Prot RT Sat Flow (s_R), veh/h/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prot RT Eff Green (g_R), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop RT Outside Lane (P_R)	0.00	0.11	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	0	1155	348	0	0	0	0	0
V/C Ratio (X)	0.00	0.47	0.85	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	0	1155	518	0	0	0	0	0
Upstream Filter (I)	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	14.6	65.6	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.4	8.8	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	15.9	74.4	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	9.1	5.6	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.4	0.4	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	9.5	6.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.08	0.33	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Intersection Summary		
HCM 6th Ctrl Delay, s/veh		21.0
HCM 6th LOS		C

Notes

User approved pedestrian interval to be less than phase max green.
User approved volume balancing among the lanes for turning movement.
User approved ignoring U-Turning movement.

FUTURE "NO-BUILD" INTERSECTION ANALYSIS

Lane Group	EBL	EBR	NBL	NBT	SBT
Lane Configurations					
Traffic Volume (vph)	67	521	157	736	959
Future Volume (vph)	67	521	157	736	959
Lane Group Flow (vph)	326	321	173	809	1080
Turn Type	Prot	Prot	pm+pt	NA	NA
Protected Phases	8	8	1	6	2
Permitted Phases			6		
Detector Phase	8	8	1	6	2
Switch Phase					
Minimum Initial (s)	6.0	6.0	5.0	15.0	15.0
Minimum Split (s)	23.5	23.5	15.0	23.5	40.5
Total Split (s)	40.0	40.0	30.0	80.0	50.0
Total Split (%)	33.3%	33.3%	25.0%	66.7%	41.7%
Yellow Time (s)	3.5	3.5	3.5	3.5	3.5
All-Red Time (s)	2.0	2.0	2.0	2.0	2.0
Lost Time Adjust (s)	0.0	0.0	0.0	0.0	0.0
Total Lost Time (s)	5.5	5.5	5.5	5.5	5.5
Lead/Lag			Lead		Lag
Lead-Lag Optimize?			Yes		Yes
Recall Mode	None	None	None	C-Min	C-Min
v/c Ratio	0.83	0.62	0.43	0.31	0.52
Control Delay (s/veh)	43.6	9.6	9.1	6.4	17.4
Queue Delay	0.0	0.0	0.0	0.0	0.0
Total Delay (s/veh)	43.6	9.6	9.1	6.4	17.4
Queue Length 50th (ft)	142	0	34	96	241
Queue Length 95th (ft)	227	76	77	172	411
Internal Link Dist (ft)	433			445	3140
Turn Bay Length (ft)	135		315		
Base Capacity (vph)	570	661	547	2617	2093
Starvation Cap Reductn	0	0	0	0	0
Spillback Cap Reductn	0	0	0	0	0
Storage Cap Reductn	0	0	0	0	0
Reduced v/c Ratio	0.57	0.49	0.32	0.31	0.52

Intersection Summary

Cycle Length: 120
Actuated Cycle Length: 120
Offset: 0 (0%), Referenced to phase 2:SBTU and 6:NBTL, Start of Green
Natural Cycle: 80
Control Type: Actuated-Coordinated

Splits and Phases: 1: Ernest W. Barrett Pkwy & Macedonia Rd

Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	67	521	157	736	0	959	24
Future Volume (veh/h)	67	521	157	736	0	959	24
Initial Q (Qb), veh	0	0	0	0		0	0
Ped-Bike Adj(A_pbT)	1.00	1.00	1.00				1.00
Parking Bus, Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	652	173	809		1054	26
Peak Hour Factor	0.91	0.91	0.91	0.91		0.91	0.91
Percent Heavy Veh, %	2	2	2	2		2	2
Cap, veh/h	410	730	368	2409		2038	50
Arrive On Green	0.00	0.23	0.06	0.68		0.58	0.58
Sat Flow, veh/h	1781	3170	1781	3647		3638	87
Grp Volume(v), veh/h	0	652	173	809		528	552
Grp Sat Flow(s),veh/h/ln	1781	1585	1781	1777		1777	1855
Q Serve(g_s), s	0.0	23.9	4.5	11.4		21.6	21.6
Cycle Q Clear(g_c), s	0.0	23.9	4.5	11.4		21.6	21.6
Prop In Lane	1.00	1.00	1.00				0.05
Lane Grp Cap(c), veh/h	410	730	368	2409		1022	1066
V/C Ratio(X)	0.00	0.89	0.47	0.34		0.52	0.52
Avail Cap(c_a), veh/h	512	911	630	2409		1022	1066
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Upstream Filter(I)	0.00	1.00	1.00	1.00		1.00	1.00
Uniform Delay (d), s/veh	0.0	44.7	11.7	8.1		15.4	15.4
Incr Delay (d2), s/veh	0.0	9.5	0.9	0.4		1.9	1.8
Initial Q Delay(d3), s/veh	0.0	0.0	0.0	0.0		0.0	0.0
%ile BackOfQ(50%),veh/ln	0.0	10.2	1.6	3.9		8.5	8.8
Unsig. Movement Delay, s/veh							
LnGrp Delay(d), s/veh	0.0	54.2	12.6	8.4		17.3	17.2
LnGrp LOS		D	B	A		B	B
Approach Vol, veh/h	652			982		1080	
Approach Delay, s/veh	54.2			9.2		17.3	
Approach LOS	D			A		B	

Timer - Assigned Phs	1	2				6		8
Phs Duration (G+Y+Rc), s	12.4	74.5				86.9		33.1
Change Period (Y+Rc), s	5.5	5.5				5.5		5.5
Max Green Setting (Gmax), s	24.5	44.5				74.5		34.5
Max Q Clear Time (g_c+I1), s	6.5	23.6				13.4		25.9
Green Ext Time (p_c), s	0.3	9.0				8.9		1.7

Intersection Summary	
HCM 6th Ctrl Delay, s/veh	23.2
HCM 6th LOS	C

Notes

User approved volume balancing among the lanes for turning movement.

User approved ignoring U-Turning movement.

Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	67	521	157	736	0	959	24
Future Volume (veh/h)	67	521	157	736	0	959	24
Number	3	18	1	6		2	12
Initial Q, veh	0	0	0	0		0	0
Ped-Bike Adj (A_pbT)	1.00	1.00	1.00				1.00
Parking Bus Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Lanes Open During Work Zone							
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	652	173	809		1054	26
Peak Hour Factor	0.91	0.91	0.91	0.91		0.91	0.91
Percent Heavy Veh, %	2	2	2	2		2	2
Opposing Right Turn Influence	Yes		Yes				
Cap, veh/h	410	730	368	2409		2038	50
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Prop Arrive On Green	0.00	0.23	0.06	0.68		0.58	0.58
Unsig. Movement Delay							
Ln Grp Delay, s/veh	0.0	54.2	12.6	8.4		17.3	17.2
Ln Grp LOS		D	B	A		B	B
Approach Vol, veh/h	652			982		1080	
Approach Delay, s/veh	54.2			9.2		17.3	
Approach LOS	D			A		B	

Timer:	1	2	3	4	5	6	7	8
Assigned Phs	1	2	8			6		
Case No	1.2	8.0	9.0			4.0		
Phs Duration (G+Y+Rc), s	12.4	74.5	33.1			86.9		
Change Period (Y+Rc), s	5.5	5.5	5.5			5.5		
Max Green (Gmax), s	24.5	44.5	34.5			74.5		
Max Allow Headway (MAH), s	3.5	5.8	3.8			5.8		
Max Q Clear (g_c+I1), s	6.5	23.6	25.9			13.4		
Green Ext Time (g_e), s	0.3	9.0	1.7			8.9		
Prob of Phs Call (p_c)	1.00	1.00	1.00			1.00		
Prob of Max Out (p_x)	0.00	0.33	0.15			0.00		
Left-Turn Movement Data								
Assigned Mvmt	1	5	3					
Mvmt Sat Flow, veh/h	1781	0	1781					
Through Movement Data								
Assigned Mvmt		2	8			6		
Mvmt Sat Flow, veh/h		3638	0			3647		
Right-Turn Movement Data								
Assigned Mvmt		12	18			16		
Mvmt Sat Flow, veh/h		87	3170			0		
Left Lane Group Data								
Assigned Mvmt	1	5	3	0	0	0	0	0
Lane Assignment	L (Pr/Pm)		L					

Lanes in Grp	1	0	1	0	0	0	0	0
Grp Vol (v), veh/h	173	0	0	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	1781	0	1781	0	0	0	0	0
Q Serve Time (g_s), s	4.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	4.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Sat Flow (s_l), veh/h/ln	522	0	1781	0	0	0	0	0
Shared LT Sat Flow (s_sh), veh/h/ln	0	0	0	0	0	0	0	0
Perm LT Eff Green (g_p), s	71.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Serve Time (g_u), s	47.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Q Serve Time (g_ps), s	11.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Time to First Blk (g_f), s	0.0	69.0	0.0	0.0	0.0	0.0	0.0	0.0
Serve Time pre Blk (g_fs), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop LT Inside Lane (P_L)	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	368	0	410	0	0	0	0	0
V/C Ratio (X)	0.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	630	0	512	0	0	0	0	0
Upstream Filter (I)	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	11.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	12.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	1.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
%ile Back of Q (50%), veh/ln	1.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Middle Lane Group Data								
Assigned Mvmt	0	2	8	0	0	6	0	0
Lane Assignment		T				T		
Lanes in Grp	0	1	0	0	0	2	0	0
Grp Vol (v), veh/h	0	528	0	0	0	809	0	0
Grp Sat Flow (s), veh/h/ln	0	1777	0	0	0	1777	0	0
Q Serve Time (g_s), s	0.0	21.6	0.0	0.0	0.0	11.4	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	21.6	0.0	0.0	0.0	11.4	0.0	0.0
Lane Grp Cap (c), veh/h	0	1022	0	0	0	2409	0	0
V/C Ratio (X)	0.00	0.52	0.00	0.00	0.00	0.34	0.00	0.00
Avail Cap (c_a), veh/h	0	1022	0	0	0	2409	0	0
Upstream Filter (I)	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	15.4	0.0	0.0	0.0	8.1	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.9	0.0	0.0	0.0	0.4	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	17.3	0.0	0.0	0.0	8.4	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	8.0	0.0	0.0	0.0	3.8	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.0	0.0	0.0	0.1	0.0	0.0

3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	8.5	0.0	0.0	0.0	3.9	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.07	0.00	0.00	0.00	0.20	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Right Lane Group Data								
Assigned Mvmt	0	12	18	0	0	16	0	0
Lane Assignment		T+R	R					
Lanes in Grp	0	1	2	0	0	0	0	0
Grp Vol (v), veh/h	0	552	652	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	0	1855	1585	0	0	0	0	0
Q Serve Time (g_s), s	0.0	21.6	23.9	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	21.6	23.9	0.0	0.0	0.0	0.0	0.0
Prot RT Sat Flow (s_R), veh/h/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prot RT Eff Green (g_R), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop RT Outside Lane (P_R)	0.00	0.05	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	0	1066	730	0	0	0	0	0
V/C Ratio (X)	0.00	0.52	0.89	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	0	1066	911	0	0	0	0	0
Upstream Filter (I)	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	15.4	44.7	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.8	9.5	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	17.2	54.2	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	8.3	9.3	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	1.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	8.8	10.2	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.07	0.56	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Intersection Summary	
HCM 6th Ctrl Delay, s/veh	23.2
HCM 6th LOS	C

Notes

User approved volume balancing among the lanes for turning movement.
User approved ignoring U-Turning movement.

Lane Group	EBL	EBR	NBL	NBT	SBT
Lane Configurations					
Traffic Volume (vph)	20	259	512	1229	952
Future Volume (vph)	20	259	512	1229	952
Lane Group Flow (vph)	150	150	551	1322	1085
Turn Type	Prot	Prot	pm+pt	NA	NA
Protected Phases	8	8	1	6	2
Permitted Phases			6		
Detector Phase	8	8	1	6	2
Switch Phase					
Minimum Initial (s)	6.0	6.0	5.0	15.0	15.0
Minimum Split (s)	23.5	23.5	15.0	23.5	40.5
Total Split (s)	30.0	30.0	45.0	120.0	75.0
Total Split (%)	20.0%	20.0%	30.0%	80.0%	50.0%
Yellow Time (s)	3.5	3.5	3.5	3.5	3.5
All-Red Time (s)	2.0	2.0	2.0	2.0	2.0
Lost Time Adjust (s)	0.0	0.0	0.0	0.0	0.0
Total Lost Time (s)	5.5	5.5	5.5	5.5	5.5
Lead/Lag			Lead		Lag
Lead-Lag Optimize?			Yes		Yes
Recall Mode	None	None	None	C-Min	C-Min
v/c Ratio	0.69	0.65	0.79	0.43	0.58
Control Delay (s/veh)	31.7	22.4	31.6	2.8	26.7
Queue Delay	0.0	0.0	0.0	0.0	0.0
Total Delay (s/veh)	31.7	22.4	31.6	2.8	26.7
Queue Length 50th (ft)	21	0	282	97	372
Queue Length 95th (ft)	94	73	454	183	504
Internal Link Dist (ft)	433			445	3140
Turn Bay Length (ft)	135		315		
Base Capacity (vph)	370	371	707	3064	1865
Starvation Cap Reductn	0	0	0	0	0
Spillback Cap Reductn	0	0	0	0	0
Storage Cap Reductn	0	0	0	0	0
Reduced v/c Ratio	0.41	0.40	0.78	0.43	0.58

Intersection Summary

Cycle Length: 150
Actuated Cycle Length: 150
Offset: 0 (0%), Referenced to phase 2:SBTU and 6:NBTL, Start of Green
Natural Cycle: 90
Control Type: Actuated-Coordinated

Splits and Phases: 1: Ernest W. Barrett Pkwy & Macedonia Rd

Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	20	259	512	1229	0	952	57
Future Volume (veh/h)	20	259	512	1229	0	952	57
Initial Q (Qb), veh	0	0	0	0		0	0
Ped-Bike Adj(A_pbT)	1.00	1.00	1.00				1.00
Parking Bus, Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	302	551	1322		1024	61
Peak Hour Factor	0.93	0.93	0.93	0.93		0.93	0.93
Percent Heavy Veh, %	2	2	2	2		2	2
Cap, veh/h	198	353	579	2897		2077	124
Arrive On Green	0.00	0.11	0.17	0.82		0.61	0.61
Sat Flow, veh/h	1781	3170	1781	3647		3501	203
Grp Volume(v), veh/h	0	302	551	1322		534	551
Grp Sat Flow(s),veh/h/ln	1781	1585	1781	1777		1777	1834
Q Serve(g_s), s	0.0	14.0	22.0	16.4		25.2	25.2
Cycle Q Clear(g_c), s	0.0	14.0	22.0	16.4		25.2	25.2
Prop In Lane	1.00	1.00	1.00				0.11
Lane Grp Cap(c), veh/h	198	353	579	2897		1083	1118
V/C Ratio(X)	0.00	0.86	0.95	0.46		0.49	0.49
Avail Cap(c_a), veh/h	291	518	747	2897		1083	1118
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Upstream Filter(I)	0.00	1.00	1.00	1.00		1.00	1.00
Uniform Delay (d), s/veh	0.0	65.5	25.8	4.1		16.3	16.3
Incr Delay (d2), s/veh	0.0	9.1	19.2	0.5		1.6	1.6
Initial Q Delay(d3), s/veh	0.0	0.0	0.0	0.0		0.0	0.0
%ile BackOfQ(50%),veh/ln	0.0	6.1	22.7	4.6		10.2	10.5
Unsig. Movement Delay, s/veh							
LnGrp Delay(d), s/veh	0.0	74.6	45.1	4.6		17.9	17.9
LnGrp LOS		E	D	A		B	B
Approach Vol, veh/h	302			1873		1085	
Approach Delay, s/veh	74.6			16.5		17.9	
Approach LOS	E			B		B	

Timer - Assigned Phs	1	2				6		8
Phs Duration (G+Y+Rc), s	30.9	96.9				127.8		22.2
Change Period (Y+Rc), s	5.5	5.5				5.5		5.5
Max Green Setting (Gmax), s	39.5	69.5				114.5		24.5
Max Q Clear Time (g_c+I1), s	24.0	27.2				18.4		16.0
Green Ext Time (p_c), s	1.4	11.8				20.8		0.7

Intersection Summary	
HCM 6th Ctrl Delay, s/veh	22.4
HCM 6th LOS	C

Notes

User approved pedestrian interval to be less than phase max green.
User approved volume balancing among the lanes for turning movement.
User approved ignoring U-Turning movement.

Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	20	259	512	1229	0	952	57
Future Volume (veh/h)	20	259	512	1229	0	952	57
Number	3	18	1	6		2	12
Initial Q, veh	0	0	0	0		0	0
Ped-Bike Adj (A_pbT)	1.00	1.00	1.00				1.00
Parking Bus Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Lanes Open During Work Zone							
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	302	551	1322		1024	61
Peak Hour Factor	0.93	0.93	0.93	0.93		0.93	0.93
Percent Heavy Veh, %	2	2	2	2		2	2
Opposing Right Turn Influence	Yes		Yes				
Cap, veh/h	198	353	579	2897		2077	124
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Prop Arrive On Green	0.00	0.11	0.17	0.82		0.61	0.61
Unsig. Movement Delay							
Ln Grp Delay, s/veh	0.0	74.6	45.1	4.6		17.9	17.9
Ln Grp LOS		E	D	A		B	B
Approach Vol, veh/h	302			1873		1085	
Approach Delay, s/veh	74.6			16.5		17.9	
Approach LOS	E			B		B	

Timer:	1	2	3	4	5	6	7	8
Assigned Phs	1	2	8			6		
Case No	1.2	8.0	9.0			4.0		
Phs Duration (G+Y+Rc), s	30.9	96.9	22.2			127.8		
Change Period (Y+Rc), s	5.5	5.5	5.5			5.5		
Max Green (Gmax), s	39.5	69.5	24.5			114.5		
Max Allow Headway (MAH), s	3.5	5.8	3.8			5.8		
Max Q Clear (g_c+I1), s	24.0	27.2	16.0			18.4		
Green Ext Time (g_e), s	1.4	11.8	0.7			20.8		
Prob of Phs Call (p_c)	1.00	1.00	1.00			1.00		
Prob of Max Out (p_x)	0.00	0.07	0.04			0.01		
Left-Turn Movement Data								
Assigned Mvmt	1	5	3					
Mvmt Sat Flow, veh/h	1781	0	1781					
Through Movement Data								
Assigned Mvmt		2	8			6		
Mvmt Sat Flow, veh/h		3501	0			3647		
Right-Turn Movement Data								
Assigned Mvmt		12	18			16		
Mvmt Sat Flow, veh/h		203	3170			0		
Left Lane Group Data								
Assigned Mvmt	1	5	3	0	0	0	0	0
Lane Assignment	L (Pr/Pm)		L					

Lanes in Grp	1	0	1	0	0	0	0	0
Grp Vol (v), veh/h	551	0	0	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	1781	0	1781	0	0	0	0	0
Q Serve Time (g_s), s	22.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	22.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Sat Flow (s_l), veh/h/ln	520	0	1781	0	0	0	0	0
Shared LT Sat Flow (s_sh), veh/h/ln	0	0	0	0	0	0	0	0
Perm LT Eff Green (g_p), s	93.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Serve Time (g_u), s	66.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Q Serve Time (g_ps), s	66.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Time to First Blk (g_f), s	0.0	91.4	0.0	0.0	0.0	0.0	0.0	0.0
Serve Time pre Blk (g_fs), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop LT Inside Lane (P_L)	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	579	0	198	0	0	0	0	0
V/C Ratio (X)	0.95	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	747	0	291	0	0	0	0	0
Upstream Filter (I)	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	25.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	19.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	45.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	19.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	3.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
%ile Back of Q (50%), veh/ln	22.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	1.83	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Middle Lane Group Data								
Assigned Mvmt	0	2	8	0	0	6	0	0
Lane Assignment		T				T		
Lanes in Grp	0	1	0	0	0	2	0	0
Grp Vol (v), veh/h	0	534	0	0	0	1322	0	0
Grp Sat Flow (s), veh/h/ln	0	1777	0	0	0	1777	0	0
Q Serve Time (g_s), s	0.0	25.2	0.0	0.0	0.0	16.4	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	25.2	0.0	0.0	0.0	16.4	0.0	0.0
Lane Grp Cap (c), veh/h	0	1083	0	0	0	2897	0	0
V/C Ratio (X)	0.00	0.49	0.00	0.00	0.00	0.46	0.00	0.00
Avail Cap (c_a), veh/h	0	1083	0	0	0	2897	0	0
Upstream Filter (I)	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	16.3	0.0	0.0	0.0	4.1	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.6	0.0	0.0	0.0	0.5	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	17.9	0.0	0.0	0.0	4.6	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	9.7	0.0	0.0	0.0	4.4	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.0	0.0	0.0	0.2	0.0	0.0

3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	10.2	0.0	0.0	0.0	4.6	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.08	0.00	0.00	0.00	0.24	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Right Lane Group Data								
Assigned Mvmt	0	12	18	0	0	16	0	0
Lane Assignment		T+R	R					
Lanes in Grp	0	1	2	0	0	0	0	0
Grp Vol (v), veh/h	0	551	302	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	0	1834	1585	0	0	0	0	0
Q Serve Time (g_s), s	0.0	25.2	14.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	25.2	14.0	0.0	0.0	0.0	0.0	0.0
Prot RT Sat Flow (s_R), veh/h/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prot RT Eff Green (g_R), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop RT Outside Lane (P_R)	0.00	0.11	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	0	1118	353	0	0	0	0	0
V/C Ratio (X)	0.00	0.49	0.86	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	0	1118	518	0	0	0	0	0
Upstream Filter (I)	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	16.3	65.5	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.6	9.1	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	17.9	74.6	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	10.0	5.7	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.4	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	10.5	6.1	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.08	0.34	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Intersection Summary		
HCM 6th Ctrl Delay, s/veh		22.4
HCM 6th LOS		C

Notes

User approved pedestrian interval to be less than phase max green.
User approved volume balancing among the lanes for turning movement.
User approved ignoring U-Turning movement.

FUTURE "BUILD" INTERSECTION ANALYSIS

Lane Group	EBL	EBR	NBL	NBT	SBU	SBT
Lane Configurations						
Traffic Volume (vph)	68	521	157	741	3	972
Future Volume (vph)	68	521	157	741	3	972
Lane Group Flow (vph)	327	321	173	814	3	1098
Turn Type	Prot	Prot	pm+pt	NA	Perm	NA
Protected Phases	8	8	1	6		2
Permitted Phases			6		2	
Detector Phase	8	8	1	6	2	2
Switch Phase						
Minimum Initial (s)	6.0	6.0	5.0	15.0	15.0	15.0
Minimum Split (s)	23.5	23.5	15.0	23.5	40.5	40.5
Total Split (s)	39.0	39.0	25.0	81.0	56.0	56.0
Total Split (%)	32.5%	32.5%	20.8%	67.5%	46.7%	46.7%
Yellow Time (s)	3.5	3.5	3.5	3.5	3.5	3.5
All-Red Time (s)	2.0	2.0	2.0	2.0	2.0	2.0
Lost Time Adjust (s)	0.0	0.0	0.0	0.0	0.0	0.0
Total Lost Time (s)	5.5	5.5	5.5	5.5	5.5	5.5
Lead/Lag			Lead		Lag	Lag
Lead-Lag Optimize?			Yes		Yes	Yes
Recall Mode	None	None	None	C-Min	C-Min	C-Min
v/c Ratio	0.83	0.61	0.45	0.31	0.01	0.52
Control Delay (s/veh)	44.1	9.5	9.5	6.5	14.7	16.8
Queue Delay	0.0	0.0	0.0	0.0	0.0	0.0
Total Delay (s/veh)	44.1	9.5	9.5	6.5	14.7	16.8
Queue Length 50th (ft)	145	0	34	97	1	240
Queue Length 95th (ft)	230	76	78	174	m7	404
Internal Link Dist (ft)	433			445		2820
Turn Bay Length (ft)	135		305		190	
Base Capacity (vph)	555	651	485	2609	384	2112
Starvation Cap Reductn	0	0	0	0	0	0
Spillback Cap Reductn	0	0	0	0	0	0
Storage Cap Reductn	0	0	0	0	0	0
Reduced v/c Ratio	0.59	0.49	0.36	0.31	0.01	0.52

Intersection Summary

Cycle Length: 120

Actuated Cycle Length: 120

Offset: 0 (0%), Referenced to phase 2:SBTU and 6:NBTL, Start of Green

Natural Cycle: 80

Control Type: Actuated-Coordinated

m Volume for 95th percentile queue is metered by upstream signal.

Splits and Phases: 1: Ernest W. Barrett Pkwy & Macedonia Rd

Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	68	521	157	741	3	972	27
Future Volume (veh/h)	68	521	157	741	3	972	27
Initial Q (Qb), veh	0	0	0	0		0	0
Ped-Bike Adj(A_pbT)	1.00	1.00	1.00				1.00
Parking Bus, Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	653	173	814		1068	30
Peak Hour Factor	0.91	0.91	0.91	0.91		0.91	0.91
Percent Heavy Veh, %	2	2	2	2		2	2
Cap, veh/h	409	729	362	2411		2033	57
Arrive On Green	0.00	0.23	0.06	0.68		0.58	0.58
Sat Flow, veh/h	1781	3170	1781	3647		3624	99
Grp Volume(v), veh/h	0	653	173	814		538	560
Grp Sat Flow(s),veh/h/ln	1781	1585	1781	1777		1777	1853
Q Serve(g_s), s	0.0	24.0	4.5	11.5		22.1	22.1
Cycle Q Clear(g_c), s	0.0	24.0	4.5	11.5		22.1	22.1
Prop In Lane	1.00	1.00	1.00				0.05
Lane Grp Cap(c), veh/h	409	729	362	2411		1023	1067
V/C Ratio(X)	0.00	0.90	0.48	0.34		0.53	0.53
Avail Cap(c_a), veh/h	497	885	551	2411		1023	1067
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Upstream Filter(I)	0.00	1.00	1.00	1.00		1.00	1.00
Uniform Delay (d), s/veh	0.0	44.8	11.8	8.0		15.5	15.5
Incr Delay (d2), s/veh	0.0	10.3	1.0	0.4		1.9	1.9
Initial Q Delay(d3), s/veh	0.0	0.0	0.0	0.0		0.0	0.0
%ile BackOfQ(50%),veh/ln	0.0	10.3	1.6	3.9		8.7	9.0
Unsig. Movement Delay, s/veh							
LnGrp Delay(d), s/veh	0.0	55.1	12.8	8.4		17.4	17.3
LnGrp LOS		E	B	A		B	B
Approach Vol, veh/h	653			987		1098	
Approach Delay, s/veh	55.1			9.2		17.4	
Approach LOS	E			A		B	

Timer - Assigned Phs	1	2				6		8
Phs Duration (G+Y+Rc), s	12.3	74.6				86.9		33.1
Change Period (Y+Rc), s	5.5	5.5				5.5		5.5
Max Green Setting (Gmax), s	19.5	50.5				75.5		33.5
Max Q Clear Time (g_c+I1), s	6.5	24.1				13.5		26.0
Green Ext Time (p_c), s	0.3	10.3				9.0		1.6

Intersection Summary	
HCM 6th Ctrl Delay, s/veh	23.4
HCM 6th LOS	C

Notes

User approved volume balancing among the lanes for turning movement.
User approved ignoring U-Turning movement.

Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	68	521	157	741	3	972	27
Future Volume (veh/h)	68	521	157	741	3	972	27
Number	3	18	1	6		2	12
Initial Q, veh	0	0	0	0		0	0
Ped-Bike Adj (A_pbT)	1.00	1.00	1.00				1.00
Parking Bus Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Lanes Open During Work Zone							
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	653	173	814		1068	30
Peak Hour Factor	0.91	0.91	0.91	0.91		0.91	0.91
Percent Heavy Veh, %	2	2	2	2		2	2
Opposing Right Turn Influence	Yes		Yes				
Cap, veh/h	409	729	362	2411		2033	57
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Prop Arrive On Green	0.00	0.23	0.06	0.68		0.58	0.58
Unsig. Movement Delay							
Ln Grp Delay, s/veh	0.0	55.1	12.8	8.4		17.4	17.3
Ln Grp LOS		E	B	A		B	B
Approach Vol, veh/h	653			987		1098	
Approach Delay, s/veh	55.1			9.2		17.4	
Approach LOS	E			A		B	

Timer:	1	2	3	4	5	6	7	8
Assigned Phs	1	2	8			6		
Case No	1.2	8.0	9.0			4.0		
Phs Duration (G+Y+Rc), s	12.3	74.6	33.1			86.9		
Change Period (Y+Rc), s	5.5	5.5	5.5			5.5		
Max Green (Gmax), s	19.5	50.5	33.5			75.5		
Max Allow Headway (MAH), s	3.5	5.8	3.8			5.8		
Max Q Clear (g_c+I1), s	6.5	24.1	26.0			13.5		
Green Ext Time (g_e), s	0.3	10.3	1.6			9.0		
Prob of Phs Call (p_c)	1.00	1.00	1.00			1.00		
Prob of Max Out (p_x)	0.00	0.24	0.23			0.00		
Left-Turn Movement Data								
Assigned Mvmt	1	5	3					
Mvmt Sat Flow, veh/h	1781	0	1781					
Through Movement Data								
Assigned Mvmt		2	8			6		
Mvmt Sat Flow, veh/h		3624	0			3647		
Right-Turn Movement Data								
Assigned Mvmt		12	18			16		
Mvmt Sat Flow, veh/h		99	3170			0		
Left Lane Group Data								
Assigned Mvmt	1	5	3	0	0	0	0	0
Lane Assignment	L (Pr/Pm)		L					

Lanes in Grp	1	0	1	0	0	0	0	0
Grp Vol (v), veh/h	173	0	0	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	1781	0	1781	0	0	0	0	0
Q Serve Time (g_s), s	4.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	4.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Sat Flow (s_l), veh/h/ln	514	0	1781	0	0	0	0	0
Shared LT Sat Flow (s_sh), veh/h/ln	0	0	0	0	0	0	0	0
Perm LT Eff Green (g_p), s	71.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Serve Time (g_u), s	47.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Q Serve Time (g_ps), s	12.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Time to First Blk (g_f), s	0.0	69.1	0.0	0.0	0.0	0.0	0.0	0.0
Serve Time pre Blk (g_fs), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop LT Inside Lane (P_L)	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	362	0	409	0	0	0	0	0
V/C Ratio (X)	0.48	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	551	0	497	0	0	0	0	0
Upstream Filter (I)	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	11.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	12.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
%ile Back of Q (50%), veh/ln	1.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Middle Lane Group Data								
Assigned Mvmt	0	2	8	0	0	6	0	0
Lane Assignment		T				T		
Lanes in Grp	0	1	0	0	0	2	0	0
Grp Vol (v), veh/h	0	538	0	0	0	814	0	0
Grp Sat Flow (s), veh/h/ln	0	1777	0	0	0	1777	0	0
Q Serve Time (g_s), s	0.0	22.1	0.0	0.0	0.0	11.5	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	22.1	0.0	0.0	0.0	11.5	0.0	0.0
Lane Grp Cap (c), veh/h	0	1023	0	0	0	2411	0	0
V/C Ratio (X)	0.00	0.53	0.00	0.00	0.00	0.34	0.00	0.00
Avail Cap (c_a), veh/h	0	1023	0	0	0	2411	0	0
Upstream Filter (I)	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	15.5	0.0	0.0	0.0	8.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.9	0.0	0.0	0.0	0.4	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	17.4	0.0	0.0	0.0	8.4	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	8.1	0.0	0.0	0.0	3.8	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.0	0.0	0.0	0.1	0.0	0.0

3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	8.7	0.0	0.0	0.0	3.9	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.08	0.00	0.00	0.00	0.20	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Right Lane Group Data								
Assigned Mvmt	0	12	18	0	0	16	0	0
Lane Assignment		T+R	R					
Lanes in Grp	0	1	2	0	0	0	0	0
Grp Vol (v), veh/h	0	560	653	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	0	1853	1585	0	0	0	0	0
Q Serve Time (g_s), s	0.0	22.1	24.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	22.1	24.0	0.0	0.0	0.0	0.0	0.0
Prot RT Sat Flow (s_R), veh/h/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prot RT Eff Green (g_R), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop RT Outside Lane (P_R)	0.00	0.05	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	0	1067	729	0	0	0	0	0
V/C Ratio (X)	0.00	0.53	0.90	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	0	1067	885	0	0	0	0	0
Upstream Filter (I)	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	15.5	44.8	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.9	10.3	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	17.3	55.1	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	8.5	9.3	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	1.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	9.0	10.3	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.08	0.57	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Intersection Summary	
HCM 6th Ctrl Delay, s/veh	23.4
HCM 6th LOS	C

Notes

User approved volume balancing among the lanes for turning movement.

User approved ignoring U-Turning movement.

Lane Group	EBL	EBR	NBL	NBT	SBU	SBT
Lane Configurations						
Traffic Volume (vph)	23	259	512	1245	8	961
Future Volume (vph)	23	259	512	1245	8	961
Lane Group Flow (vph)	153	150	551	1339	9	1096
Turn Type	Prot	Prot	pm+pt	NA	Perm	NA
Protected Phases	8	8	1	6		2
Permitted Phases			6		2	
Detector Phase	8	8	1	6	2	2
Switch Phase						
Minimum Initial (s)	6.0	6.0	5.0	15.0	15.0	15.0
Minimum Split (s)	23.5	23.5	15.0	23.5	40.5	40.5
Total Split (s)	28.0	28.0	55.0	122.0	67.0	67.0
Total Split (%)	18.7%	18.7%	36.7%	81.3%	44.7%	44.7%
Yellow Time (s)	3.5	3.5	3.5	3.5	3.5	3.5
All-Red Time (s)	2.0	2.0	2.0	2.0	2.0	2.0
Lost Time Adjust (s)	0.0	0.0	0.0	0.0	0.0	0.0
Total Lost Time (s)	5.5	5.5	5.5	5.5	5.5	5.5
Lead/Lag			Lead		Lag	Lag
Lead-Lag Optimize?			Yes		Yes	Yes
Recall Mode	None	None	None	C-Min	C-Min	C-Min
v/c Ratio	0.70	0.64	0.79	0.44	0.05	0.59
Control Delay (s/veh)	32.5	22.0	31.9	2.9	24.1	28.0
Queue Delay	0.0	0.0	0.0	0.0	0.0	0.0
Total Delay (s/veh)	32.5	22.0	31.9	2.9	24.1	28.0
Queue Length 50th (ft)	24	0	294	102	4	372
Queue Length 95th (ft)	97	73	434	190	m18	564
Internal Link Dist (ft)	433			445		2820
Turn Bay Length (ft)	135		305		190	
Base Capacity (vph)	351	353	752	3059	199	1848
Starvation Cap Reductn	0	0	0	0	0	0
Spillback Cap Reductn	0	0	0	0	0	0
Storage Cap Reductn	0	0	0	0	0	0
Reduced v/c Ratio	0.44	0.42	0.73	0.44	0.05	0.59

Intersection Summary

Cycle Length: 150

Actuated Cycle Length: 150

Offset: 0 (0%), Referenced to phase 2:SBTU and 6:NBTL, Start of Green

Natural Cycle: 90

Control Type: Actuated-Coordinated

m Volume for 95th percentile queue is metered by upstream signal.

Splits and Phases: 1: Ernest W. Barrett Pkwy & Macedonia Rd

Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	23	259	512	1245	8	961	59
Future Volume (veh/h)	23	259	512	1245	8	961	59
Initial Q (Qb), veh	0	0	0	0		0	0
Ped-Bike Adj(A_pbT)	1.00	1.00	1.00				1.00
Parking Bus, Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	305	551	1339		1033	63
Peak Hour Factor	0.93	0.93	0.93	0.93		0.93	0.93
Percent Heavy Veh, %	2	2	2	2		2	2
Cap, veh/h	199	354	580	2896		2057	125
Arrive On Green	0.00	0.11	0.17	0.81		0.60	0.60
Sat Flow, veh/h	1781	3170	1781	3647		3496	207
Grp Volume(v), veh/h	0	305	551	1339		539	557
Grp Sat Flow(s),veh/h/ln	1781	1585	1781	1777		1777	1833
Q Serve(g_s), s	0.0	14.2	22.5	16.8		25.9	25.9
Cycle Q Clear(g_c), s	0.0	14.2	22.5	16.8		25.9	25.9
Prop In Lane	1.00	1.00	1.00				0.11
Lane Grp Cap(c), veh/h	199	354	580	2896		1074	1108
V/C Ratio(X)	0.00	0.86	0.95	0.46		0.50	0.50
Avail Cap(c_a), veh/h	267	476	858	2896		1074	1108
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Upstream Filter(I)	0.00	1.00	1.00	1.00		1.00	1.00
Uniform Delay (d), s/veh	0.0	65.5	26.6	4.1		16.8	16.8
Incr Delay (d2), s/veh	0.0	11.7	15.4	0.5		1.7	1.6
Initial Q Delay(d3), s/veh	0.0	0.0	0.0	0.0		0.0	0.0
%ile BackOfQ(50%),veh/ln	0.0	6.3	22.1	4.7		10.5	10.8
Unsig. Movement Delay, s/veh							
LnGrp Delay(d), s/veh	0.0	77.2	42.0	4.7		18.5	18.5
LnGrp LOS		E	D	A		B	B
Approach Vol, veh/h	305			1890		1096	
Approach Delay, s/veh	77.2			15.5		18.5	
Approach LOS	E			B		B	

Timer - Assigned Phs	1	2				6		8
Phs Duration (G+Y+Rc), s	31.6	96.2				127.7		22.3
Change Period (Y+Rc), s	5.5	5.5				5.5		5.5
Max Green Setting (Gmax), s	49.5	61.5				116.5		22.5
Max Q Clear Time (g_c+I1), s	24.5	27.9				18.8		16.2
Green Ext Time (p_c), s	1.5	11.3				21.4		0.6

Intersection Summary			
HCM 6th Ctrl Delay, s/veh			22.2
HCM 6th LOS			C

Notes

User approved volume balancing among the lanes for turning movement.
User approved ignoring U-Turning movement.

Movement	EBL	EBR	NBL	NBT	SBU	SBT	SBR
Lane Configurations							
Traffic Volume (veh/h)	23	259	512	1245	8	961	59
Future Volume (veh/h)	23	259	512	1245	8	961	59
Number	3	18	1	6		2	12
Initial Q, veh	0	0	0	0		0	0
Ped-Bike Adj (A_pbT)	1.00	1.00	1.00				1.00
Parking Bus Adj	1.00	1.00	1.00	1.00		1.00	1.00
Work Zone On Approach	No			No		No	
Lanes Open During Work Zone							
Adj Sat Flow, veh/h/ln	1870	1870	1870	1870		1870	1870
Adj Flow Rate, veh/h	0	305	551	1339		1033	63
Peak Hour Factor	0.93	0.93	0.93	0.93		0.93	0.93
Percent Heavy Veh, %	2	2	2	2		2	2
Opposing Right Turn Influence	Yes		Yes				
Cap, veh/h	199	354	580	2896		2057	125
HCM Platoon Ratio	1.00	1.00	1.00	1.00		1.00	1.00
Prop Arrive On Green	0.00	0.11	0.17	0.81		0.60	0.60
Unsig. Movement Delay							
Ln Grp Delay, s/veh	0.0	77.2	42.0	4.7		18.5	18.5
Ln Grp LOS		E	D	A		B	B
Approach Vol, veh/h	305			1890		1096	
Approach Delay, s/veh	77.2			15.5		18.5	
Approach LOS	E			B		B	

Timer:	1	2	3	4	5	6	7	8
Assigned Phs	1	2	8			6		
Case No	1.2	8.0	9.0			4.0		
Phs Duration (G+Y+Rc), s	31.6	96.2	22.3			127.7		
Change Period (Y+Rc), s	5.5	5.5	5.5			5.5		
Max Green (Gmax), s	49.5	61.5	22.5			116.5		
Max Allow Headway (MAH), s	3.5	5.8	3.8			5.8		
Max Q Clear (g_c+I1), s	24.5	27.9	16.2			18.8		
Green Ext Time (g_e), s	1.5	11.3	0.6			21.4		
Prob of Phs Call (p_c)	1.00	1.00	1.00			1.00		
Prob of Max Out (p_x)	0.00	0.14	0.17			0.01		
Left-Turn Movement Data								
Assigned Mvmt	1	5	3					
Mvmt Sat Flow, veh/h	1781	0	1781					
Through Movement Data								
Assigned Mvmt		2	8			6		
Mvmt Sat Flow, veh/h		3496	0			3647		
Right-Turn Movement Data								
Assigned Mvmt		12	18			16		
Mvmt Sat Flow, veh/h		207	3170			0		
Left Lane Group Data								
Assigned Mvmt	1	5	3	0	0	0	0	0
Lane Assignment	L (Pr/Pm)		L					

Lanes in Grp	1	0	1	0	0	0	0	0
Grp Vol (v), veh/h	551	0	0	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	1781	0	1781	0	0	0	0	0
Q Serve Time (g_s), s	22.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	22.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Sat Flow (s_l), veh/h/ln	515	0	1781	0	0	0	0	0
Shared LT Sat Flow (s_sh), veh/h/ln	0	0	0	0	0	0	0	0
Perm LT Eff Green (g_p), s	92.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Serve Time (g_u), s	64.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Perm LT Q Serve Time (g_ps), s	64.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Time to First Blk (g_f), s	0.0	90.7	0.0	0.0	0.0	0.0	0.0	0.0
Serve Time pre Blk (g_fs), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop LT Inside Lane (P_L)	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	580	0	199	0	0	0	0	0
V/C Ratio (X)	0.95	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	858	0	267	0	0	0	0	0
Upstream Filter (I)	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	26.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	15.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	42.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	19.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	2.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
%ile Back of Q (50%), veh/ln	22.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	1.84	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Middle Lane Group Data								
Assigned Mvmt	0	2	8	0	0	6	0	0
Lane Assignment		T				T		
Lanes in Grp	0	1	0	0	0	2	0	0
Grp Vol (v), veh/h	0	539	0	0	0	1339	0	0
Grp Sat Flow (s), veh/h/ln	0	1777	0	0	0	1777	0	0
Q Serve Time (g_s), s	0.0	25.9	0.0	0.0	0.0	16.8	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	25.9	0.0	0.0	0.0	16.8	0.0	0.0
Lane Grp Cap (c), veh/h	0	1074	0	0	0	2896	0	0
V/C Ratio (X)	0.00	0.50	0.00	0.00	0.00	0.46	0.00	0.00
Avail Cap (c_a), veh/h	0	1074	0	0	0	2896	0	0
Upstream Filter (I)	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	16.8	0.0	0.0	0.0	4.1	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.7	0.0	0.0	0.0	0.5	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	18.5	0.0	0.0	0.0	4.7	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	10.0	0.0	0.0	0.0	4.4	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.0	0.0	0.0	0.2	0.0	0.0

3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	10.5	0.0	0.0	0.0	4.7	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.09	0.00	0.00	0.00	0.24	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Right Lane Group Data								
Assigned Mvmt	0	12	18	0	0	16	0	0
Lane Assignment		T+R	R					
Lanes in Grp	0	1	2	0	0	0	0	0
Grp Vol (v), veh/h	0	557	305	0	0	0	0	0
Grp Sat Flow (s), veh/h/ln	0	1833	1585	0	0	0	0	0
Q Serve Time (g_s), s	0.0	25.9	14.2	0.0	0.0	0.0	0.0	0.0
Cycle Q Clear Time (g_c), s	0.0	25.9	14.2	0.0	0.0	0.0	0.0	0.0
Prot RT Sat Flow (s_R), veh/h/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prot RT Eff Green (g_R), s	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prop RT Outside Lane (P_R)	0.00	0.11	1.00	0.00	0.00	0.00	0.00	0.00
Lane Grp Cap (c), veh/h	0	1108	354	0	0	0	0	0
V/C Ratio (X)	0.00	0.50	0.86	0.00	0.00	0.00	0.00	0.00
Avail Cap (c_a), veh/h	0	1108	476	0	0	0	0	0
Upstream Filter (I)	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00
Uniform Delay (d1), s/veh	0.0	16.8	65.5	0.0	0.0	0.0	0.0	0.0
Incr Delay (d2), s/veh	0.0	1.6	11.7	0.0	0.0	0.0	0.0	0.0
Initial Q Delay (d3), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Control Delay (d), s/veh	0.0	18.5	77.2	0.0	0.0	0.0	0.0	0.0
1st-Term Q (Q1), veh/ln	0.0	10.3	5.7	0.0	0.0	0.0	0.0	0.0
2nd-Term Q (Q2), veh/ln	0.0	0.5	0.6	0.0	0.0	0.0	0.0	0.0
3rd-Term Q (Q3), veh/ln	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
%ile Back of Q Factor (f_B%)	0.00	1.00	1.00	0.00	0.00	1.00	0.00	0.00
%ile Back of Q (50%), veh/ln	0.0	10.8	6.3	0.0	0.0	0.0	0.0	0.0
%ile Storage Ratio (RQ%)	0.00	0.10	0.35	0.00	0.00	0.00	0.00	0.00
Initial Q (Qb), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Final (Residual) Q (Qe), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Delay (ds), s/veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Q (Qs), veh	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sat Cap (cs), veh/h	0	0	0	0	0	0	0	0
Initial Q Clear Time (tc), h	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Intersection Summary	
HCM 6th Ctrl Delay, s/veh	22.2
HCM 6th LOS	C

Notes

User approved volume balancing among the lanes for turning movement.
User approved ignoring U-Turning movement.

Traffic Volume Worksheets

25-196 Res Dev at Old Horseshoe Bend Rd and Ernest Barrett Pkwy - Cobb County, GA

Traffic Volumes

A&R Engineering
November 2025

1.Ernest W.B.Pkwy @ Macedonia

A.M. Peak Hour

	Ernest W. Barrett Parkway Northbound					Ernest W. Barrett Parkway Southbound					Macedonia Road Eastbound					- Westbound				
Condition	U	L	T	R	Tot	U	L	T	R	Tot	U	L	T	R	Tot	U	L	T	R	Tot
Existing 2025 Traffic Counts:	0	154	722	0	876	0	0	940	24	964	0	66	0	511	577	0	0	0	0	0
Growth Factor (%):	1	1	1	1		1	1	1	1		1	1	1	1		1	1	1	1	
No-Build 2027 Volumes:	0	157	736	0	893	0	0	959	24	983	0	67	0	521	588	0	0	0	0	0
Total New Trips:	0	0	5	0	5	3	0	13	3	19	0	1	0	0	1	0	0	0	0	0
Future 2027 Traffic Volumes:	0	157	741	0	898	3	0	972	27	1002	0	68	0	521	589	0	0	0	0	0

P.M. Peak Hour

	Ernest W. Barrett Parkway Northbound					Ernest W. Barrett Parkway Southbound					Macedonia Road Eastbound					- Westbound				
Condition	U	L	T	R	Tot	U	L	T	R	Tot	U	L	T	R	Tot	U	L	T	R	Tot
Existing 2025 Traffic Counts:	0	502	1205	0	1707	0	0	933	56	989	0	20	0	254	274	0	0	0	0	0
Growth Factor (%):	1	1	1	1		1	1	1	1		1	1	1	1		1	1	1	1	
No-Build 2027 Volumes:	0	512	1229	0	1741	0	0	952	57	1009	0	20	0	259	279	0	0	0	0	0
Total New Trips:	0	0	16	0	16	8	0	9	2	19	0	3	0	0	3	0	0	0	0	0
Future 2027 Traffic Volumes:	0	512	1245	0	1757	8	0	961	59	1028	0	23	0	259	282	0	0	0	0	0

25-196 Res Dev at Old Horseshoe Bend Rd and Ernest Barrett Pkwy - Cobb County, GA

Traffic Volumes

A&R Engineering
November 2025

2.Ernest W.B.Pkwy @ Median Brk

A.M. Peak Hour

	Ernest W. Barrett Parkway Northbound					Ernest W. Barrett Parkway Southbound					- Eastbound					- Westbound				
Condition	U	L	T	R	Tot	U	L	T	R	Tot	U	L	T	R	Tot	U	L	T	R	Tot
Existing 2025 Traffic Counts:	1	0	787	0	788	3	0	964	0	967	0	0	0	0	0	0	0	0	0	0
Growth Factor (%):	1	1	1	1		1	1	1	1		1	1	1	1		1	1	1	1	
No-Build 2027 Volumes:	1	0	803	0	804	3	0	983	0	986	0	0	0	0	0	0	0	0	0	0
Total New Trips:	15	0	10	0	25	0	0	3	0	3	0	0	0	0	0	0	0	0	0	0
Future 2027 Traffic Volumes:	16	0	813	0	829	3	0	986	0	989	0	0	0	0	0	0	0	0	0	0

P.M. Peak Hour

	Ernest W. Barrett Parkway Northbound					Ernest W. Barrett Parkway Southbound					- Eastbound					- Westbound				
Condition	U	L	T	R	Tot	U	L	T	R	Tot	U	L	T	R	Tot	U	L	T	R	Tot
Existing 2025 Traffic Counts:	10	0	1305	0	1315	1	0	951	0	952	0	0	0	0	0	0	0	0	0	0
Growth Factor (%):	1	1	1	1		1	1	1	1		1	1	1	1		1	1	1	1	
No-Build 2027 Volumes:	10	0	1331	0	1341	1	0	970	0	971	0	0	0	0	0	0	0	0	0	0
Total New Trips:	10	0	7	0	17	0	0	8	0	8	0	0	0	0	0	0	0	0	0	0
Future 2027 Traffic Volumes:	20	0	1338	0	1358	1	0	978	0	979	0	0	0	0	0	0	0	0	0	0

25-196 Res Dev at Old Horseshoe Bend Rd and Ernest Barrett Pkwy - Cobb County, GA

Traffic Volumes

A&R Engineering
November 2025

3.Ernest W.B.Pkwy @ Drwy (RIRO)

A.M. Peak Hour

	Ernest W. Barrett Parkway Northbound					Ernest W. Barrett Parkway Southbound					- Eastbound					Site Driveway (Right-In/ Right-Out) Westbound				
Condition	U	L	T	R	Tot	U	L	T	R	Tot	U	L	T	R	Tot	U	L	T	R	Tot
Existing 2025 Traffic Counts:	0	0	788	0	788	0	0	965	0	965	0	0	0	0	0	0	0	0	0	0
Growth Factor (%):	1	1	1	1		1	1	1	1		1	1	1	1		1	1	1	1	
No-Build 2027 Volumes:	0	0	804	0	804	0	0	984	0	984	0	0	0	0	0	0	0	0	0	0
Total New Trips:	0	0	0	9	9	0	0	18	0	18	0	0	0	0	0	0	0	0	25	25
Future 2027 Traffic Volumes:	0	0	804	9	813	0	0	1002	0	1002	0	0	0	0	0	0	0	0	25	25

P.M. Peak Hour

	Ernest W. Barrett Parkway Northbound					Ernest W. Barrett Parkway Southbound					- Eastbound					Site Driveway (Right-In/ Right-Out) Westbound				
Condition	U	L	T	R	Tot	U	L	T	R	Tot	U	L	T	R	Tot	U	L	T	R	Tot
Existing 2025 Traffic Counts:	0	0	1315	0	1315	0	0	961	0	961	0	0	0	0	0	0	0	0	0	0
Growth Factor (%):	1	1	1	1		1	1	1	1		1	1	1	1		1	1	1	1	
No-Build 2027 Volumes:	0	0	1341	0	1341	0	0	980	0	980	0	0	0	0	0	0	0	0	0	0
Total New Trips:	0	0	0	27	27	0	0	18	0	18	0	0	0	0	0	0	0	0	17	17
Future 2027 Traffic Volumes:	0	0	1341	27	1368	0	0	998	0	998	0	0	0	0	0	0	0	0	17	17